

JUST Undergraduate Final Admission System

A Project Report Submitted

In partial fulfillment of the requirements for the degree of
B.Sc. Engineering in Computer Science and Engineering



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Preface

On the eve of the beginning of civilization, human beings had been striving tirelessly to discover, explore, and innovate unknown aspects of the world. This inherent curiosity and desire for advancement have driven humanity toward continuous progress and development. Over time, human beings have not only acquired theoretical knowledge but have also emphasized the importance of practical implementation, which plays a crucial role in solving real-world problems and improving the quality of life.

In the modern era, the field of Computer Science and Engineering has emerged as one of the most dynamic and influential disciplines, contributing significantly to technological advancement and digital transformation. The philosophy of the Computer Science and Engineering department is centered on equipping students with both theoretical understanding and practical skills. To achieve this goal, the department has introduced thesis and project programs that encourage students to engage in real-life problem-solving and innovative system development.

As a part of fulfilling the academic requirements of the undergraduate program, I have undertaken a project titled **JUST Undergraduate Final Admission System**. This project aims to develop a comprehensive web-based platform that simplifies and streamlines the admission process for undergraduate students. It focuses on efficiency, transparency, and user-friendliness, ensuring a better experience for both applicants and administrators.

Throughout the development of this project, I have conducted extensive research, analysis, and observation to understand the existing admission processes and identify their limitations. This has enabled me to design and implement a system that addresses real-world challenges effectively. The project has provided me with valuable opportunities to enhance my technical skills, problem-solving abilities, and understanding of software development methodologies.

I believe that this project will not only serve as an academic accomplishment but also contribute positively to the automation and digitalization of admission systems. The knowledge and experience gained during this work will undoubtedly play a vital role in my future career and professional development.

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Chapter 1

Introduction

1.1 Background

Admission of a university is an indispensable activity in every educational activity and is as old as education itself. With more students applying for admission than in previous years, admission process for admissions officers has become increasingly time-consuming. More applications resulted in heavier paperwork and processing challenges. Every admission is routed through various faculty for evaluation, and the manual admission process caused difficulty in admission, and archiving.

In order to speed up and simplify the admission process, the development of the Automated Final Admission System is to enact document redesign and an automated management program. The proposed “JUST undergraduate final admission system” would store, route, and retrieve prospective and current application documents. The development of science and technology has immensely contributed to the growth of computers and the internet which has adversely increased the use and need for an automated applicant’s admission system.

Universities, Colleges, Companies, organizations, etc. are seeking better ways to distribute and disseminate information throughout the world. In order to achieve this, they develop and rely on online applications, which help them input, update, process, and disseminate information. The internet has increased the use of online applications because data could be easily and readily accessed as well as documented. This project is targeted at allowing admission of applicants’ decisions to be made faster and more efficiently. It looks at improving the present traditional system of the manual process (paper-based) which is manually documented. This type of project is a hot topic in the whole world to digitalize our education system.

1.2 Motivation

1.2.1 Existing System

There are several automated systems used for managing final admission processes in educational institutions. These systems differ in features, capabilities, and implementation depending on institutional needs and available resources.

Currently, JUST provides a web-based application titled “JUST Admission 2025”. While this system includes some useful functionalities, it is not a fully integrated or complete solution. Several important features are missing, such as admission confirmation, sending admission confirmation messages, and verification of required documents.

Another issue in the current system is that any user can register regardless of whether they have been selected for admission or not. This creates inconsistency and reduces system reliability. Additionally, the system lacks proper validation mechanisms, which makes it difficult to ensure the authenticity and correctness of submitted data.

1.2.2 Limitations of the System

The current system suffers from several limitations that affect its efficiency and usability.

First, there is no proper validation process implemented. As a result, users can input incorrect or incomplete data without restriction. Second, the system allows unrestricted data insertion but does not provide proper mechanisms for updating or removing data, which can lead to data inconsistency.

Third, the system architecture and code structure are not well organized, making it difficult to maintain, scale, or extend the system in the future.

Another significant limitation is that users who are not eligible (i.e., those who did not get a chance for admission) can still register in the system. This reduces the accuracy and trustworthiness of the admission process.

Finally, key features such as admission confirmation, automated message sending, and document verification are either missing or not fully implemented, which prevents the system from functioning as a complete automated admission solution.

1.3 Objectives

The primary objective of this project is to develop a user-friendly and reliable web-based admission system that ensures maximum accuracy and efficiency. The system aims to provide the following facilities:

- **Efficiency:** Automate the admission process to reduce manual effort, save time, and streamline data handling.
- **Accuracy:** Minimize errors by reducing manual data entry and ensuring proper handling of application data and documents.
- **Consistency:** Ensure that all data and applications are processed uniformly, reducing the risk of inconsistency and data loss.
- **Validation and Reliability:** Implement proper validation mechanisms to ensure that only eligible candidates can register and submit accurate information.
- **Improved Performance:** Enhance system performance and reliability by using modern technologies and optimized system architecture.
- **Improved Security:** Strengthen system security by incorporating updated security measures to protect user data and prevent unauthorized access.
- **Cost Effectiveness:** Reduce operational costs by minimizing the need for manual labor and paper-based processes.
- **Objectivity:** Ensure a fair and unbiased admission process by evaluating applicants based on predefined and objective criteria.

Chapter 2

Software Requirement Specification

2.1 Introduction

2.1.1 Purpose

The primary purpose of the JUST Undergraduate Final Admission System (JUST-UFAS) is to provide a secure, efficient, and user-friendly platform for completing the final admission process. The previous manual and semi-automated system had several limitations, including lack of proper validation, poor code quality, absence of structured design patterns, and no mechanism to enforce admission deadlines.

This system is the first structured and fully integrated solution developed by the JUST Undergraduate Admission Committee. It aims to eliminate existing issues by introducing proper validation, deadline control, role-based access, and a more maintainable and scalable architecture.

This Software Requirement Specification (SRS) document describes the complete workflow of the final admission process, including:

- Applicant interaction through online application form submission.
- Verification and clearance by the Dean Office with required documents.
- Bank receipt verification by the Hall Office.
- Medical examination and reporting by the Medical Center.
- Final approval and document collection by the Registrar Office.

- Administrative control for managing users, roles, and system operations.

2.1.2 Document Conventions

The following conventions are used throughout this SRS document:

- Section headings are numbered hierarchically.
- **Bold text** indicates important terms or concepts.
- All figures and tables are labeled with appropriate names and numbering.
- The following abbreviations are used:
 - JUST: Jashore University of Science and Technology.
 - SRS: Software Requirements Specification.
 - JUST-UFAS: JUST Undergraduate Final Admission System.

2.1.3 Intended Audience and Reading Suggestions

This document follows IEEE SRS standards and is intended for:

- Admission Committee members
- Project Supervisor
- Chairman of the CSE Department
- Internal and External Viva Board Members
- Future Developers and System Maintainers
- Any stakeholders interested in understanding the system

Readers are encouraged to begin with the Introduction and Overall Description sections before proceeding to detailed requirements.

2.2 Overall Description

2.2.1 Product Perspective

The JUST-UFAS is a web-based automated admission management system designed to replace the existing manual and partially digital processes. The previous system lacked critical features such as proper validation mechanisms, deadline enforcement, and structured system architecture.

The proposed system introduces:

- Strong input validation to prevent invalid or incomplete submissions.
- Deadline control to restrict registration and updates after the admission deadline.
- Role-based access control to ensure secure and authorized operations.
- A modular and maintainable architecture following modern design patterns.
- Real-time tracking of application progress across different offices.
- Scalability to handle a large number of concurrent applicants.

2.2.2 Product Functions

The major functionalities of the JUST-UFAS are as follows:

- **Admin Module:**
 - Manage users and roles.
 - Control system operations (open/close admission).
 - Monitor overall admission progress.
- **Dean Office Module:**
 - Verify documents and validate applicants.
 - Approve or reject applications.
 - Export applicant data.
- **Hall Office Module:**
 - Verify bank receipts.

- Manage residential clearance.
- **Medical Center Module:**
 - Record medical data.
 - Generate reports and provide clearance.
- **Registrar Office Module:**
 - Final admission approval.
 - Document collection and archiving.
- **Applicant Module:**
 - Registration and profile management.
 - Application form submission.
 - Document upload and download.

2.3 User Classes and Characteristics

The system includes multiple user roles, each with specific responsibilities. All users are expected to have basic knowledge of computers and web browsers.

Admin:

- Full system control.
- Manage users and roles.
- Control admission deadlines.

Dean Office:

- Verify documents.
- Approve or reject applications.

Hall Office:

- Verify payment and clearance.

Medical Office:

- Generate medical reports.

Registrar Office:

- Final admission approval.

Applicant:

- Submit application.
- Upload documents.
- Track status.

2.4 Constraints

The system operates under the following constraints:

- Admission is strictly controlled by deadlines.
- No registration or updates allowed after deadline.
- All inputs must pass validation rules.
- Role-based access control is enforced.
- Requires stable internet connection.
- Security measures must protect user data.

2.5 External Interface Requirements

2.5.1 User Interfaces

The system provides a web-based user interface designed for simplicity and usability. Interfaces are role-based, ensuring that each user (Admin, Dean Office, Hall Office, Medical Office, Registrar Office, Applicant) can only access permitted features.

The interface includes:

- Clear navigation and dashboard
- Form validation with proper instructions
- Role-specific panels for efficient task completion

2.5.2 Integration with Existing Data

The system supports integration with pre-existing admission data provided by the administration in Excel format. The admin uploads this data, which is then used to generate applicant accounts.

2.5.3 Integration with Internal Modules

All modules (Dean, Hall, Medical, Registrar) are interconnected and share a centralized database to ensure real-time updates and consistency.

2.5.4 Authentication and Authorization

The system implements secure authentication and role-based authorization:

- Admin provides login credentials to users
- Applicants log in using GST Application ID
- Access control ensures users can only perform authorized actions

2.6 System Features: Admin Management

2.6.1 Description and Priority

The Admin module has the highest priority and controls the entire system. The admin uploads applicant data (Excel file), manages users, controls admission deadlines, and monitors all activities.

2.6.2 Functional Requirements

Authentication

- Input: Username and password
- Output: Secure login access

Upload Applicant Data

- Input: Excel file containing applicant information
- Output: System generates applicant accounts

Manage Users

- View, update, and delete users

Update Applicant Information

- Modify applicant data if required

Control Admission

- Open or close admission system

View Reports

- Monitor admission progress and seat status

2.7 System Features: Dean Office

2.7.1 Description

The Dean Office verifies applicant information and approves or rejects applications based on eligibility criteria.

2.7.2 Functional Requirements

- Search applicant using ID
- View applicant profile
- Verify documents
- Approve or reject applicant
- View applicant lists (all, pending, approved)

2.8 System Features: Hall Office

2.8.1 Description

The Hall Office verifies payment and residential clearance.

2.8.2 Functional Requirements

- Search applicant
- View profile
- Verify bank receipt
- Approve applicant

2.9 System Features: Medical Office

2.9.1 Description

The Medical Office handles physical examination and medical clearance.

2.9.2 Functional Requirements

- Search applicant
- Insert medical data
- Generate report
- Approve applicant

2.10 System Features: Registrar Office

2.10.1 Description

The Registrar Office gives final admission approval and stores documents.

2.10.2 Functional Requirements

- Search applicant
- Receive documents
- Final approval
- Generate reports

2.11 System Features: Applicant

2.11.1 Description

Applicants do not register manually. Their accounts are created by admin-uploaded data. Applicants log in, complete their profile, and download the application form.

2.11.2 Functional Requirements

- Login using provided credentials
- Complete application form
- Upload required documents
- Download application form

2.12 Non-Functional Requirements

2.12.1 Performance

The system should handle a large number of users efficiently.

2.12.2 Security

Sensitive data must be protected using authentication and authorization.

2.12.3 Scalability

The system should support increasing users without performance loss.

2.12.4 Maintainability

The system should be easy to update and maintain.

2.12.5 Usability

The system must be user-friendly and easy to operate.

2.12.6 Reliability

The system should ensure high availability and minimal downtime.

Chapter 3

System Design and Methodology

3.1 Use Case Diagram

3.1.1 User Management

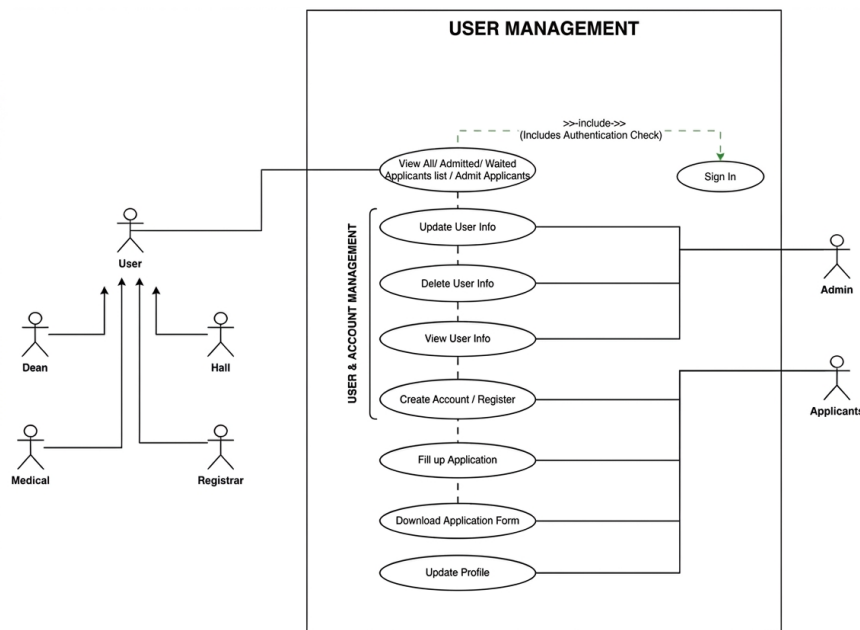


Figure 3.1: Use Case Diagram of User Management

The User Management use case diagram consists of six primary actors: Admin, Dean Office, Hall Office, Medical Office, Registrar Office, and Applicant.

3.1.1.1 Sign in

This use case describes the authentication process for the *JUST Undergraduate Final Admission System*. It ensures that only authorized personnel and registered applicants can access the system's restricted functionalities.

Characteristics	Description
Use Case ID	UC-1.1
Use Case Name	Sign in
Primary Actors	Admin, Dean Office, Hall Office, Medical Office, Registrar Office, Applicant
Goal in Context	To gain authorized access to the system dashboard.
Precondition	The user must have a valid account with a Username and Password.
Post-condition	The user is redirected to their respective dashboard based on their role.
Triggers	User attempts to access a protected route or clicks the Sign-in link.
Scenario	1. The user navigates to the System Login page. 2. The user enters their Username/Email and Password. 3. The user clicks the Login button. 4. The Next.js backend validates the credentials against the PostgreSQL database using Prisma. 5. Upon successful validation, a secure session is established. 6. The user is granted access to the system dashboard.
Exception	Invalid Credentials: If the credentials do not match, the system displays an "Access Denied" error message.
Frequency of Use	Extremely High (Daily).

Table 3.1: Use Case Table: Sign in (UC-1.1)

3.1.1.2 View Applicant List

This use case describes the procedure for administrative actors to access and review the records of candidates who have applied for admission. It allows various administrative offices (Dean, Hall, Medical, Registrar) to filter and monitor the status of

applicants throughout the admission lifecycle.

Characteristics	Description
Use Case ID	UC-1.2
Use Case Name	View Applicant List
Primary Actors	Admin, Dean Office, Hall Office, Medical Office, Registrar Office
Goal in Context	To filter and view different applicant categories (All, Admitted, or Waiting).
Precondition	The user must be securely authenticated and signed into the administrative dashboard.
Post-condition	The requested list of applicants is fetched and displayed on the UI.
Triggers	Need to verify candidate status or proceed with departmental clearance.
Scenario	1. The user visits the Dashboard after a successful login. 2. The user clicks on the 'Applicant Management' or 'View List' option from the sidebar. 3. The user chooses a specific list category (e.g., All Applicants, Admitted, or Waiting). 4. The Next.js backend fetches the data from the PostgreSQL database through Prisma. 5. The system displays the applicant records in a responsive data table.
Exception	Access Denied: If the user's role does not have permission to view certain lists, the system shows an unauthorized access message.
Frequency of Use	High (Used frequently for administrative tasks).

Table 3.2: Use Case Table: View Applicant List (UC-1.2)

3.1.1.3 Update User Information

This use case describes the administrative process of modifying existing user data within the system. It ensures that the Admin can maintain accurate records for all institutional roles in the *JUST Undergraduate Final Admission System*, allowing for corrections in personal details, roles, or account status.

Characteristics	Description
Use Case ID	UC-1.3
Use Case Name	Update User Information
Primary Actors	Admin
Goal in Context	To modify, correct, or update the profile information of any registered user.
Precondition	Admin must be securely authenticated and signed into the administrative dashboard.
Post-condition	The updated user information is successfully persisted in the PostgreSQL database.
Triggers	Requirement to update contact details, institutional roles, or personal information of a user.
Scenario	1. The Admin navigates to the User Management section of the dashboard. 2. The Admin searches for the specific user and clicks on the Edit icon/button. 3. The system populates a form with the user's current data from the database. 4. The Admin modifies the necessary fields (e.g., Name, Email, Role, or Status). 5. The Admin clicks the Update or Save Changes button. 6. The Next.js backend performs validation and updates the record via Prisma. 7. A success notification is displayed to the Admin.
Exception	Validation Error: If a duplicate email or invalid data format is entered, the system prevents the update and shows an error message.
Frequency of Use	High (Used for regular data maintenance).

Table 3.3: Use Case Table: Update User Information (UC-1.3)

3.1.1.4 Delete User Information

This use case describes the administrative process of permanently removing a user's record from the *JUST Undergraduate Final Admission System*. It allows the Admin to maintain system integrity by deleting redundant, unauthorized, or incorrect accounts from the database.

Characteristics	Description
Use Case ID	UC-1.4
Use Case Name	Delete User Information
Primary Actors	Admin
Goal in Context	To permanently remove a specific user record from the system database.
Precondition	The Admin must be signed in and authorized with administrative privileges.
Post-condition	The user record is successfully removed from the PostgreSQL database via Prisma.
Triggers	Admin identifies an account that needs to be removed for maintenance or security reasons.
Scenario	1. The Admin navigates to the User Management section in the dashboard. 2. The system fetches the list of all registered users from the database. 3. The Admin identifies and selects the specific user to be deleted. 4. The Admin clicks the Delete button associated with the user. 5. A confirmation modal appears to prevent accidental deletion. 6. Upon confirmation, the backend executes the delete query using Prisma. 7. The system updates the UI and displays a success notification.
Exception	Deletion Failed: If the user is linked to critical active records, the system prevents deletion and shows a "Constraint Error."
Frequency of Use	Low to Medium (As needed for system cleanup).

Table 3.4: Use Case Table: Delete User Information (UC-1.4)

3.1.1.5 View User Information

This use case allows the Admin to view detailed profiles and access levels of all users (Applicants and Staff) registered within the system. It is essential for verifying

user identity, checking account status, and managing system-wide permissions in the *JUST Undergraduate Final Admission System*.

Characteristics	Description
Use Case ID	UC-1.5
Use Case Name	View User Information
Primary Actors	Admin
Goal in Context	To display the full profile details, account status, and role of a specific user.
Precondition	Admin must be authenticated and signed into the administrative dashboard.
Post-condition	Detailed user information is retrieved from the database and displayed on the screen.
Triggers	Admin needs to verify user credentials or investigate a specific account's activity.
Scenario	1. The Admin navigates to the User Management or All Users section in the dashboard. 2. The Admin uses the search bar to find a specific user by their ID, Name, or Email. 3. The Admin clicks on the 'View Details' or 'Profile' icon of the selected user. 4. The Next.js API fetches the specific user's record from the PostgreSQL database via Prisma. 5. The system renders a detailed profile view including personal, contact, and role information.
Exception	User Not Found: If the requested ID does not exist, the system displays a "Record Not Found" error message.
Frequency of Use	High (Used frequently for administrative oversight).

Table 3.5: Use Case Table: View User Information (UC-1.5)

3.1.1.6 Fill-up Application Form

This use case outlines the process for a registered applicant to submit their academic and personal details through the *JUST Undergraduate Final Admission System*. It

is a multi-step procedure designed to collect necessary documentation and information required for the final admission validation.

Characteristics	Description
Use Case ID	UC-1.6
Use Case Name	Fill-up Application Form
Primary Actors	Applicant
Goal in Context	To successfully submit the complete admission application form.
Precondition	The applicant must be signed in and should not have a previously submitted pending application.
Post-condition	The application data is stored in the database, and the status is set to "Pending" for office review.
Triggers	Applicant clicks on the "Apply Now" or "Fill-up Form" button on their dashboard.
Scenario	1. The applicant navigates to the application form section. 2. The system presents a multi-step form (Personal Info, Academic History, Document Upload). 3. The applicant enters all required information and uploads necessary files (Photo, Signature, etc.). 4. The applicant reviews the information for any errors. 5. The applicant clicks the Submit button. 6. The Next.js frontend sends the data to the API, which persists it in PostgreSQL via Prisma. 7. A success confirmation message is displayed to the applicant.
Exception	Validation Error: If required fields are empty or files are in an unsupported format, the system prompts the user to correct the errors.
Frequency of Use	Once per applicant (High volume during admission periods).

Table 3.6: Use Case Table: Fill-up Application Form (UC-1.6)

3.1.1.7 Download Application Form

This use case provides the applicant with the ability to generate and download a formal PDF version of their submitted application. It serves as an official record for the applicant, which is essential for the physical verification process at *Jashore University of Science and Technology (JUST)*.

Characteristics	Description
Use Case ID	UC-1.7
Use Case Name	Download Application Form
Primary Actors	Applicant
Goal in Context	To obtain a digital or physical copy of the completed admission form for official documentation.
Precondition	The applicant must be securely signed in and must have successfully submitted their admission application.
Post-condition	A PDF document is dynamically generated and successfully downloaded to the applicant's local device.
Triggers	The applicant clicks the Download PDF or Print Application button on their personal dashboard.
Scenario	1. The applicant navigates to their personal dashboard. 2. The system verifies the application status from the PostgreSQL database. 3. The applicant clicks the Download PDF button. 4. The Next.js server-side logic triggers the PDF generation engine using the applicant's stored data. 5. The system initiates the file download stream in the user's browser. 6. The applicant saves the generated PDF file for future reference and physical submission.
Exception	Generation Error: If there is a server-side failure or the database is unreachable, a "PDF Generation Failed" error is displayed.
Frequency of Use	High (Once per applicant after the successful submission of the form).

Table 3.7: Use Case Table: Download Application Form (UC-1.7)

3.1.2 Dean Office Management

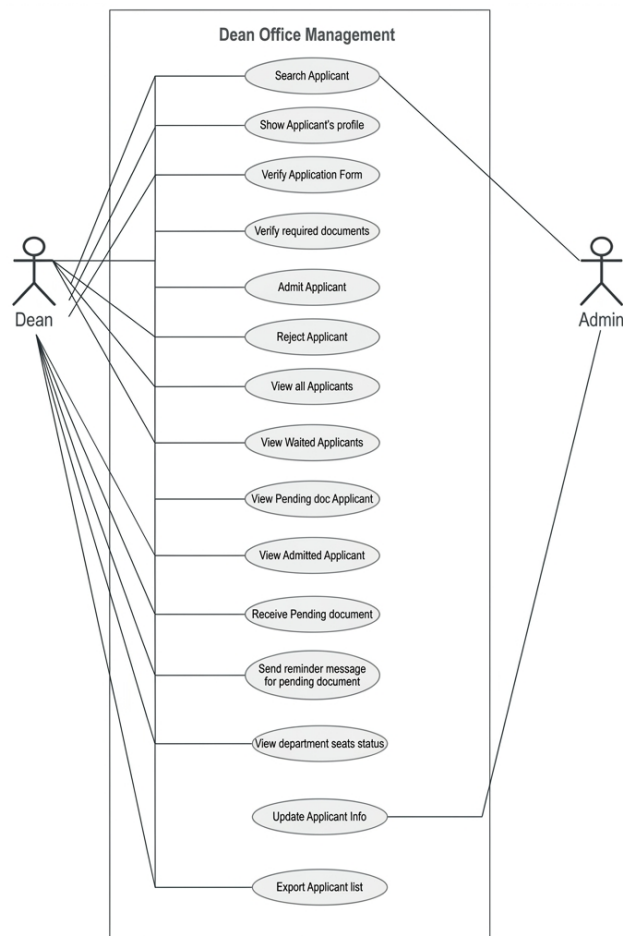


Figure 3.2: Use Case Diagram of Dean Office Management

The Dean Office Management use case diagram consists of two primary actors: Admin and Dean Office. This module controls various types of use cases including searching applicants, viewing applicant profiles, verifying application forms, verifying required documents, admitting applicants, rejecting applicants, and managing applicant lists. It also provides functionalities to view all applicants, waited applicants, and admitted applicants with or without pending documents.

3.1.2.1 Search Applicant

This use case describes how the Dean Office searches for an applicant.

Characteristics	Description
Use Case ID	UC-2.1
Use Case Name	Search Applicant
Primary Actors	Dean Office
Goal in Context	To search an applicant using ID or roll.
Precondition	Must be signed in and have Applicant ID or Roll.
Triggers	Need to search a specific applicant.
Scenario	1. Visit the Dean Office dashboard. 2. Enter Applicant ID or Roll in search box. 3. Press Enter.
Exception	Invalid role.
Frequency of Use	High

Table 3.8: Use Case Table: Search Applicant (UC-2.1)

3.1.2.2 Show Applicant Profile

This use case describes how the Dean Office views an individual applicant's profile. It provides a detailed view of the candidate's personal information and academic records, which is essential for making informed admission decisions.

Characteristics	Description
Use Case ID	UC-2.2
Use Case Name	Show Applicant Profile
Primary Actors	Dean Office
Goal in Context	To display the detailed profile of a specific applicant for review.
Precondition	The user must be signed in with Dean Office credentials and have searched for an applicant.
Post-condition	The system successfully displays the applicant's detailed data from the PostgreSQL database via Prisma ORM.
Triggers	Need to review specific academic or personal details of a candidate for admission processing.
Scenario	1. The user searches for a specific applicant from the applicant list. 2. The user clicks on the Profile or View Details button. 3. The system fetches the applicant data using Prisma. 4. The system displays the comprehensive profile including personal details, academic results, and payment records.
Exception	Data Not Found: If the record does not exist, an error message is shown. Access Denied: Unauthorized users are restricted.
Frequency of Use	High (During the admission screening phase).

Table 3.9: Use Case Table: Show Applicant Profile (UC-2.2)

3.1.2.3 Verify Application Form

This use case describes the procedure for the Dean Office to verify the submitted application form. It ensures that the applicant's data and uploaded documents are consistent with the university's requirements before granting final approval.

Characteristics	Description
Use Case ID	UC-2.3
Use Case Name	Verify Application Form
Primary Actors	Dean Office
Goal in Context	To successfully validate and approve the applicant's submitted application form.
Precondition	The user must be signed in with Dean Office privileges and an application must be pending review.
Post-condition	The application status is updated to 'Verified' in the PostgreSQL database and reflected on the system.
Triggers	Requirement to process and authorize candidate applications for the final admission list.
Scenario	1. The user searches for a specific applicant record from the pending list. 2. The user navigates to the Verification or Review section. 3. The system displays the submitted data and documents via the interface. 4. The user reviews the information and clicks the Approve or Verify button. 5. The system updates the status using Prisma ORM.
Exception	Data Mismatch: If the information is incorrect, the user may reject the form or request a correction from the applicant.
Frequency of Use	High (During peak admission cycles).

Table 3.10: Use Case Table: Verify Application Form (UC-2.3)

3.1.2.4 Verify Required Documents

This use case describes how the Dean Office verifies the physical or uploaded documents of an applicant. It ensures all academic certificates and required papers are authentic and meet the university's criteria.

Characteristics	Description
Use Case ID	UC-2.4
Use Case Name	Verify Required Documents
Primary Actors	Dean Office
Goal in Context	To successfully verify and confirm the validity of applicant documents.
Precondition	The user must be signed in and the applicant's record must be accessed.
Post-condition	The document status is updated to 'Verified' in the PostgreSQL database.
Triggers	Initiation of document screening during the admission process.
Scenario	1. Search for a specific applicant from the dashboard. 2. Click on the Verify Required Documents section. 3. Select the specific documents provided by the candidate. 4. Click the Verify button to confirm via Prisma ORM.
Exception	Invalid Role: Unauthorized staff attempt verification. Missing Docs: System flags missing files.
Frequency of Use	High

Table 3.11: Use Case Table: Verify Required Documents (UC-2.4)

3.1.2.5 Admit Applicant (Dean Clearance)

This use case allows the Dean Office to grant final clearance for admission after successful verification of all records.

Characteristics	Description
Use Case ID	UC-2.5
Use Case Name	Admit Applicant
Primary Actors	Dean Office
Goal in Context	To grant final admission clearance to the applicant.
Precondition	All documents must be verified and the applicant must be eligible.
Post-condition	Applicant status is changed to 'Admitted' and department seat count is updated.
Scenario	1. Search for an eligible applicant record. 2. Review the final summary of the candidate. 3. Click the Admit button. 4. System updates the record in PostgreSQL.
Exception	Seat capacity full or role unauthorized.
Frequency of Use	High

Table 3.12: Use Case Table: Admit Applicant (UC-2.5)

3.1.2.6 Reject Applicant

This use case describes the procedure for rejecting an applicant if significant discrepancies are found in their documents or eligibility.

Characteristics	Description
Use Case ID	UC-2.6
Use Case Name	Reject Applicant
Primary Actors	Dean Office
Goal in Context	To disqualify an applicant due to errors or non-compliance.
Scenario	1. Search for the specific applicant. 2. Click the Reject button. 3. Provide a reason for rejection (optional/required). 4. System updates the status to 'Rejected'.
Exception	Invalid role or accidental rejection of verified candidates.
Frequency of Use	High

Table 3.13: Use Case Table: Reject Applicant (UC-2.6)

3.1.2.7 View Applicant Lists (All, Waited, Admitted)

These use cases describe how the Dean Office monitors different categories of applicants through the dashboard sidebar.

Characteristics	Description
Use Case ID	UC-2.7, 2.8, 2.9
Use Case Name	View All / Waited / Admitted Applicants
Primary Actors	Dean Office
Scenario	1. Log in to the Dean Dashboard. 2. Select the respective item (All / Waited / Admitted) from the sidebar. 3. The system queries the PostgreSQL database via Prisma and displays the filtered list.
Frequency of Use	High

Table 3.14: Use Case Table: Applicant List Management (UC-2.7 - 2.9)

3.1.2.8 Pending Documents and Reminders

This section covers receiving pending documents and sending reminder messages to applicants who have yet to complete their documentation.

Characteristics	Description
Use Case ID	UC-2.11, 2.12, 2.13
Use Case Name	Manage Pending Documents & Send Reminders
Primary Actors	Dean Office
Scenario (Reminder)	1. Identify applicant with missing documents. 2. Select the missing items and click Send Message. 3. System triggers a notification to the applicant's contact.
Scenario (Receive)	1. Click Pending Documents Action from the sidebar. 2. Update the record upon receiving physical copies.

Table 3.15: Use Case Table: Pending Documents Management (UC-2.11 - 2.13)

3.1.2.9 Seats Status and Data Export

Characteristics	Description
Use Case ID	UC-2.14, 2.15
Use Case Name	View Seats Status & Export Lists
Primary Actors	Dean
Goal in Context	To monitor enrollment capacity and generate reports in CSV/Excel format.
Scenario (Export)	1. Navigate to the desired applicant list. 2. Click the Export button. 3. The system generates and downloads the file.

Table 3.16: Use Case Table: Monitoring and Export (UC-2.14 - 2.15)

3.1.3 Hall Office Management

The Hall Office Management use case diagram consists of two primary actors: Admin and Hall Authority. This module handles various operations including searching applicants, admitting them after bank receipt verification, managing lists, and updating information.

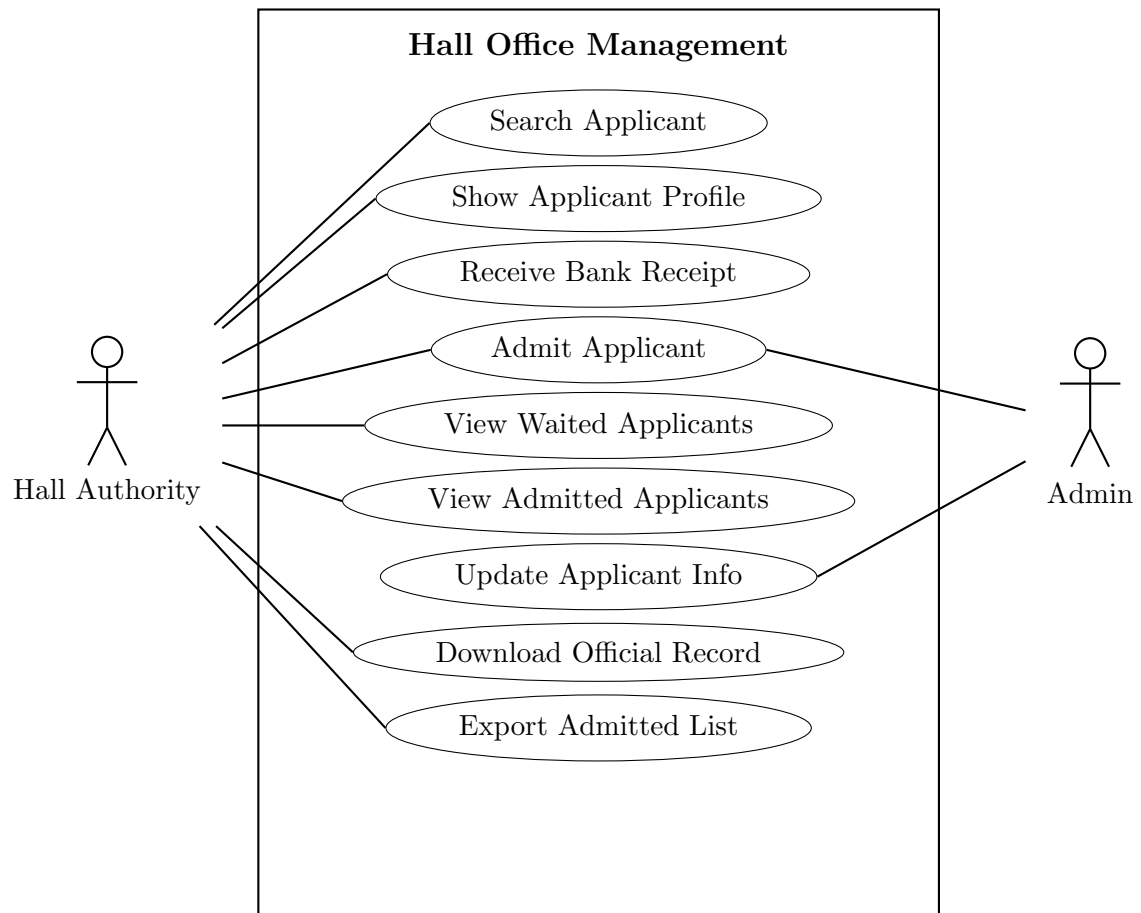


Figure 3.3: Use Case Diagram of Hall Office Management

3.1.3.1 Search Applicant

This use case describes how the Hall Office searches for an applicant within the system using specific identifiers.

Characteristics	Description
Use Case ID	UC-3.1
Use Case Name	Search Applicant
Primary Actors	Hall Office
Goal in Context	To find a specific applicant's record in the system.
Precondition	Must be signed in and have GST Applicant ID or Roll.
Scenario	1. User navigates to the Hall Office dashboard. 2. User enters ID/Roll in the search field. 3. User submits the search query.
Frequency of Use	High

Table 3.17: Use Case Table: Search Applicant (UC-3.1)

3.1.3.2 Show Applicant Profile

Characteristics	Description
Use Case ID	UC-3.2
Use Case Name	Show Applicant Profile
Primary Actors	Hall Office
Goal	Display detailed academic and personal information.
Scenario	1. Search applicant. 2. Click 'Profile' button.

Table 3.18: Use Case: Show Applicant Profile (UC-3.2)

3.1.3.3 Receive Bank Receipt

Characteristics	Description
Use Case ID	UC-3.3
Use Case Name	Receive Bank Receipt
Primary Actors	Hall Office
Goal	Confirm the physical or digital bank receipt for hall fees.
Scenario	1. Search applicant. 2. Click 'Receive Receipt' button. 3. System updates PostgreSQL via Prisma.

Table 3.19: Use Case: Receive Bank Receipt (UC-3.3)

3.1.3.4 Admit Applicant (Hall Clearance)

Characteristics	Description
Use Case ID	UC-3.4
Use Case Name	Admit Applicant (Hall Clearance)
Primary Actors	Hall Office
Goal	Grant clearance for final admission.
Scenario	1. Verify receipt. 2. Click 'Admit' button.

Table 3.20: Use Case: Admit Applicant (UC-3.4)

3.1.3.5 View Applicant Lists

Characteristics	Description
Use Case ID	UC-3.5, 3.6
Use Case Name	View Waited / Admitted Applicants
Primary Actors	Hall Office
Goal	Monitor the status of candidates in different phases.
Scenario	1. Navigate to sidebar. 2. Select respective list (Waited/Admitted).

Table 3.21: Use Case: List Management (UC-3.5 - 3.6)

3.1.3.6 Update Applicant Info

Characteristics	Description
Use Case ID	UC-3.7
Use Case Name	Update Applicant Information
Primary Actors	Admin
Goal	Correct or update data for an applicant.
Scenario	1. Click Applicant table. 2. Edit values. 3. Click Update.

Table 3.22: Use Case: Update Applicant Info (UC-3.7)

3.1.3.7 Export and Download Records

Characteristics	Description
Use Case ID	UC-3.8, 3.9
Use Case Name	Export List / Download Official Record
Primary Actors	Hall Office
Goal	Generate PDF or CSV reports for office records.
Scenario	1. Go to list. 2. Click Export/PDF button.

Table 3.23: Use Case: Reporting (UC-3.8 - 3.9)

3.1.4 Medical Office Management

The Medical Office Management module handles the physical examination and medical clearance of applicants. This use case diagram involves two primary actors: Medical Officer and Admin. The system allows managing medical records, generating reports, and granting final medical clearance.

3.1.4.1 Search Applicant

Characteristics	Description
Use Case ID	UC-4.1
Use Case Name	Search Applicant
Primary Actors	Medical Office
Scenario	1. Visit Dashboard. 2. Input Search box. 3. Press Enter.

Table 3.24: Use Case Table: Search Applicant (UC-4.1)

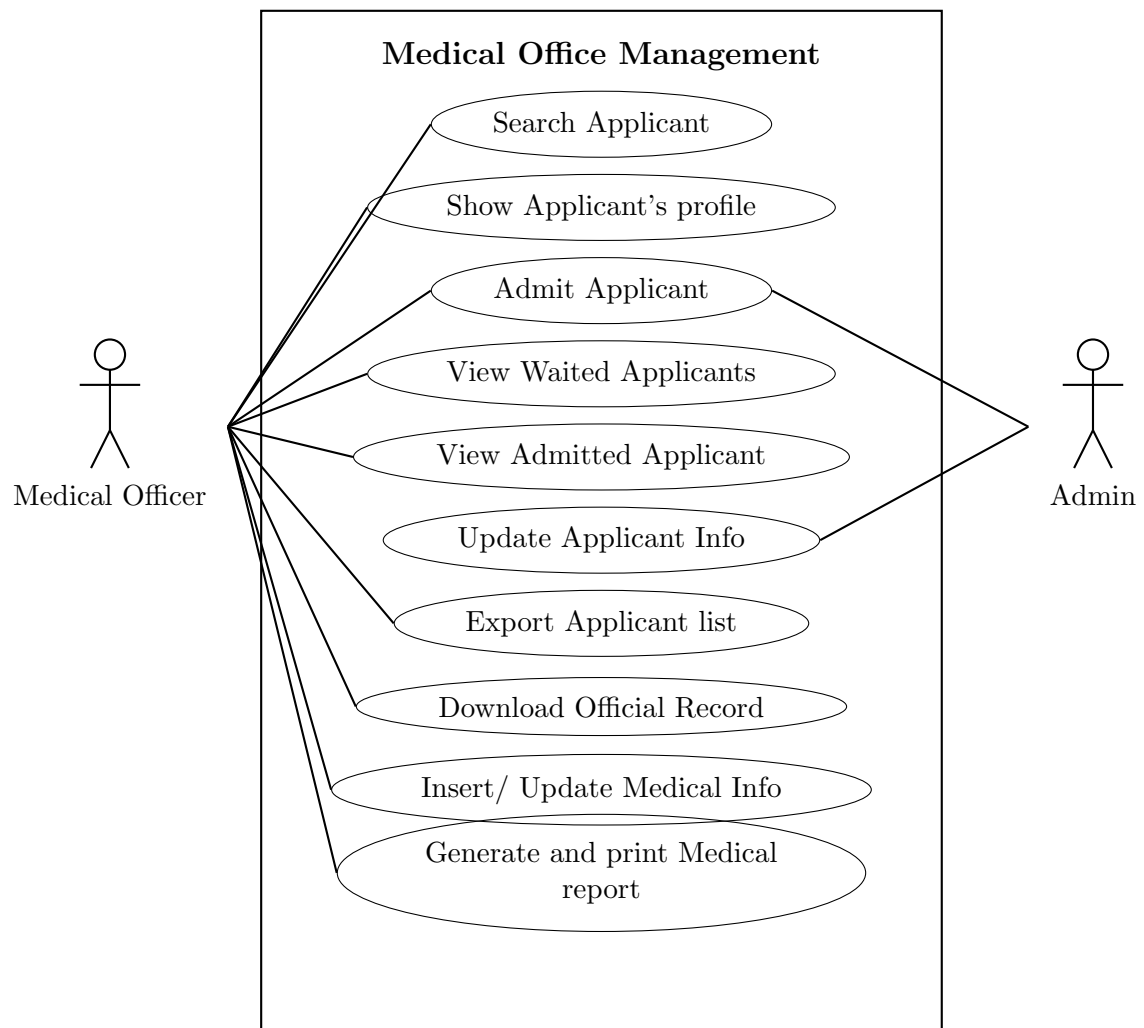


Figure 3.4: Use Case Diagram of Medical Office Management

3.1.4.2 Show Applicant Profile

Characteristics	Description
Use Case ID	UC-4.2
Use Case Name	Show Applicant Profile
Primary Actors	Medical Office
Scenario	1. Search Applicant. 2. Click on Profile button.

Table 3.25: Use Case Table: Show Applicant Profile (UC-4.2)

3.1.4.3 Insert/ Update Medical Info

Characteristics	Description
Use Case ID	UC-4.3
Use Case Name	Insert/ Update Medical Info
Primary Actors	Medical Office
Goal	To Examine Applicant's physical condition and record data.
Scenario	1. Search Applicant. 2. Input medical info. 3. Click submit to save.

Table 3.26: Use Case Table: Insert/ Update Medical Info (UC-4.3)

3.1.4.4 Generate and Print Medical Report

Characteristics	Description
Use Case ID	UC-4.4
Use Case Name	Generate and Print Medical Report
Primary Actors	Medical Office
Scenario	1. Search Applicant. 2. Click Generate and Print button.

Table 3.27: Use Case Table: Generate and Print Medical Report (UC-4.4)

3.1.4.5 Admit Applicant (Medical Clearance)

Characteristics	Description
Use Case ID	UC-4.5
Use Case Name	Admit Applicant
Primary Actors	Medical Office
Goal	Grant clearance after physical examination.
Scenario	1. Search Applicant. 2. Click Admit Button.

Table 3.28: Use Case Table: Admit Applicant (UC-4.5)

3.1.4.6 View Applicant Lists

Characteristics	Description
Use Case ID	UC-4.6, 4.7
Use Case Name	View Waited / Admitted Applicant
Primary Actors	Medical Office
Scenario	1. Log in. 2. Click sidebar list item.

Table 3.29: Use Case Table: View Applicant Lists (UC-4.6 - 4.7)

3.1.4.7 Update Applicant Info

Characteristics	Description
Use Case ID	UC-4.9
Use Case Name	Update Applicant Information
Primary Actors	Admin
Scenario	1. Go to update page. 2. Input values. 3. Click update.

Table 3.30: Use Case Table: Update Applicant Info (UC-4.9)

3.1.4.8 Export and Download Records

Characteristics	Description
Use Case ID	UC-4.8, 4.10, 4.11
Use Case Name	Export List / Download Official Record
Primary Actors	Medical Office
Scenario	1. Log in. 2. Click Export/PDF button from list.

Table 3.31: Use Case Table: Reporting (UC-4.8 - 4.11)

3.1.5 Registrar Office Management

The Registrar Office Management use case diagram consists of two primary actors: Admin and Registrar Office. This module controls operations including searching applicants, viewing profiles, receiving documents, and monitoring seat status.

3.1.5.1 Search Applicant

Characteristics	Description
Use Case ID	UC-5.1
Use Case Name	Search Applicant
Primary Actors	Registrar Office
Goal	Quickly locate specific applicant records within the database using unique identifiers.
Precondition	User must be signed in with Registrar Office credentials.
Scenario	1. Visit the dashboard. 2. Navigate to the search bar or applicant list. 3. Input Applicant ID or Roll Number. 4. Press 'Enter' or click the search icon to display results.
Exception	Unauthorized access or record not found.
Frequency of Use	High frequency across all administrative modules.

Table 3.32: Use Case Table: Search Applicant (UC-5.1)

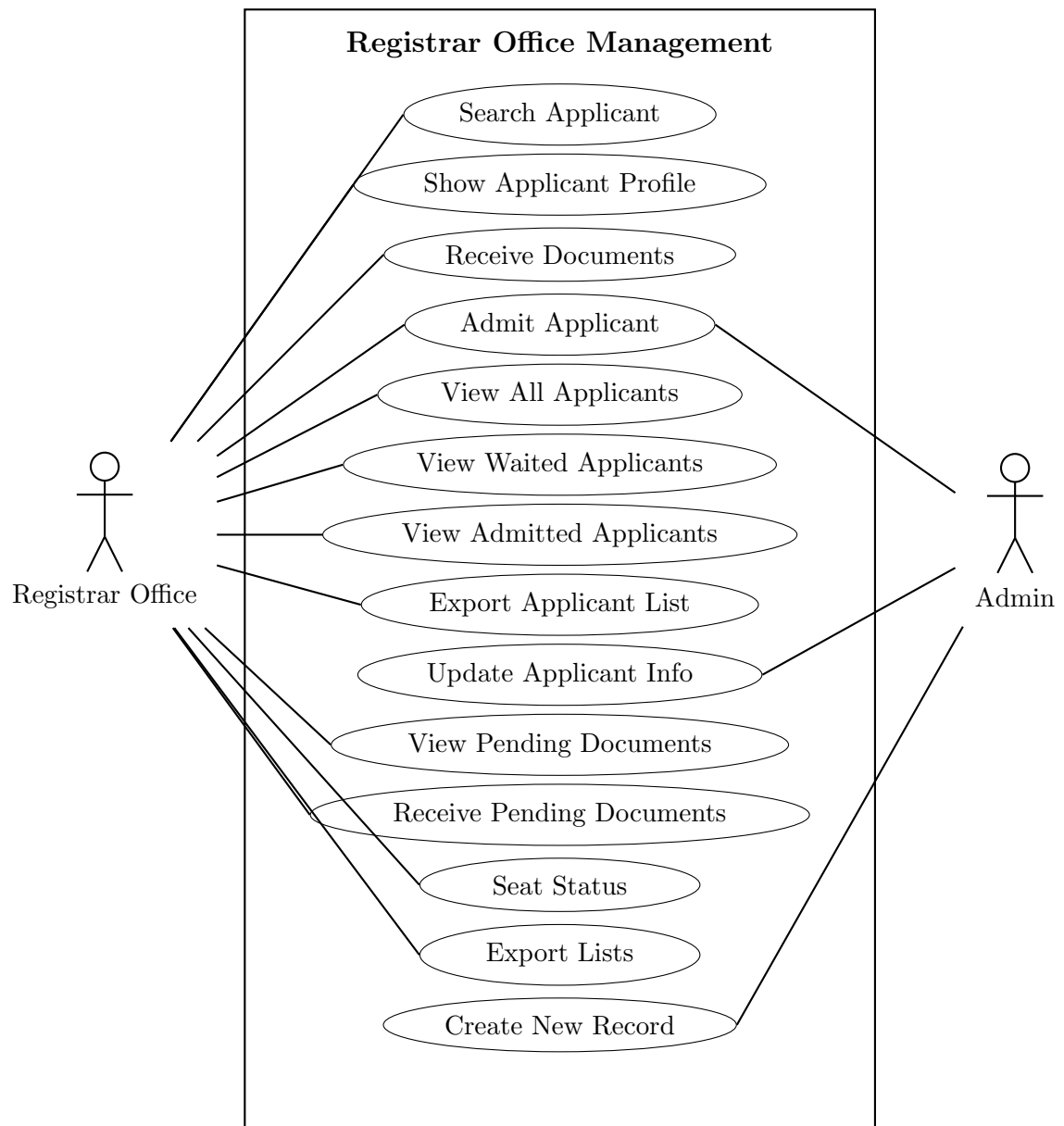


Figure 3.5: Use Case Diagram of Registrar Office Management

3.1.5.2 Show Applicant Profile

Characteristics	Description
Use Case ID	UC-5.2
Use Case Name	Show Applicant Profile
Primary Actors	Registrar Office
Goal	View the complete personal and academic profile of a specific applicant.
Precondition	User must search for a specific applicant from the database.
Scenario	1. Search for the applicant. 2. Click on the 'View Profile' or applicant name link. 3. System displays the detailed profile page.
Exception	Unauthorized access or record not found.
Frequency of Use	High frequency during the data verification and admission process.

Table 3.33: Use Case Table: Show Applicant Profile (UC-5.2)

3.1.5.3 Receive Documents

Characteristics	Description
Use Case ID	UC-5.3
Use Case Name	Receive Documents
Primary Actors	Registrar Office
Goal	Formally record and confirm the physical or digital submission of required documents.
Precondition	Applicant must be searched and selected from the database.
Scenario	1. Search for the specific applicant. 2. Select the applicant from the result list. 3. Click the 'Receive Documents' button to update the status.
Exception	Missing mandatory documents or unauthorized user role.
Frequency of Use	High frequency during the document verification stage.

Table 3.34: Use Case Table: Receive Documents (UC-5.3)

3.1.5.4 Admit Applicant

Characteristics	Description
Use Case ID	UC-5.4
Use Case Name	Admit Applicant
Primary Actors	Registrar Office
Goal	Finalize the admission process for an eligible applicant.
Precondition	Applicant documents must be received and verified by the registrar.
Scenario	1. Search for the specific applicant. 2. Verify eligibility status. 3. Click the 'Admit' button to confirm admission in the system.
Frequency of Use	High frequency during admission sessions.

Table 3.35: Use Case Table: Admit Applicant (UC-5.4)

3.1.5.5 View All Applicant

Characteristics	Description
Use Case ID	UC-5.5
Use Case Name	View All Applicant
Primary Actors	Registrar Office
Goal	Provide a comprehensive overview of all registered applicants within the system.
Precondition	User must be signed in with Registrar Office credentials.
Scenario	1. Log in to the dashboard. 2. Navigate to the sidebar menu. 3. Click on 'All Applicants' to view the complete list and their current status.
Exception	Unauthorized access or session timeout.
Frequency of Use	High frequency during the entire admission cycle.

Table 3.36: Use Case Table: View All Applicant (UC-5.5)

3.1.5.6 View Waited Applicant

Characteristics	Description
Use Case ID	UC-5.6
Use Case Name	View Waited Applicant
Primary Actors	Registrar Office
Goal	Monitor and manage the list of applicants who are currently on the waiting list for admission.
Precondition	User must be signed in with Registrar Office credentials.
Scenario	1. Log in to the dashboard. 2. Navigate to the sidebar menu. 3. Click on 'Waited Applicant' to view the prioritized waiting list.
Exception	Unauthorized access or session timeout.
Frequency of Use	High frequency during the seat allocation and admission finalization stages.

Table 3.37: Use Case Table: View Waited Applicant (UC-5.6)

3.1.5.7 View Admitted Applicant

Characteristics	Description
Use Case ID	UC-5.7
Use Case Name	View Admitted Applicant
Primary Actors	Registrar Office
Goal	Access and monitor the list of all students who have successfully completed admission.
Precondition	User must be signed in with Registrar Office credentials.
Scenario	1. Log in to the dashboard. 2. Navigate to the sidebar menu. 3. Click on 'Admitted Applicant' to view the full list.
Exception	Unauthorized access or session timeout.
Frequency of Use	High frequency during and after the admission period.

Table 3.38: Use Case Table: View Admitted Applicant (UC-5.7)

3.1.5.8 Export Bulk Applicant Lists

Characteristics	Description
Use Case ID	UC-5.13
Use Case Name	Export Bulk Applicant Lists
Primary Actors	Registrar
Goal	Export comprehensive lists of all applicants for high-level administrative reporting.
Precondition	User must be signed in with Registrar-level administrative privileges.
Scenario	1. Log in to the dashboard. 2. Navigate to the master applicant list. 3. Click the 'Export' button to generate files for all applicants.
Exception	Unauthorized user role or server-side timeout during bulk processing.
Frequency of Use	High frequency during the final stages of admission cycles.

Table 3.39: Use Case Table: Export Bulk Applicant Lists (UC-5.13)

3.1.5.9 Update Applicant Information

Characteristics	Description
Use Case ID	UC-5.9
Use Case Name	Update Applicant Information
Primary Actors	Admin
Goal	Manually update or correct specific fields of an applicant's profile to maintain data accuracy.
Precondition	Admin must be authenticated and authorized to access the applicant records.
Scenario	1. Log in to the dashboard. 2. Navigate to the Applicant table. 3. Access the update page for the specific applicant. 4. Input corrected values. 5. Click the 'Update' button to save changes.
Exception	Unauthorized access or database validation error.
Frequency of Use	High frequency during the data verification and correction phase.

Table 3.40: Use Case Table: Update Applicant Information (UC-5.9)

3.1.5.10 View Pending Document Applicant

Characteristics	Description
Use Case ID	UC-5.10
Use Case Name	View Pending Document Applicant
Primary Actors	Registrar Office
Goal	Access and review the list of applicants who have not yet completed their document submission or verification.
Precondition	User must be signed in with Registrar Office credentials.
Scenario	1. Log in to the dashboard. 2. Navigate to the sidebar menu. 3. Click on 'Documents Pending' to view the specific applicant list.
Exception	Unauthorized access or session timeout.
Frequency of Use	High frequency during the document verification phase.

Table 3.41: Use Case Table: View Pending Document Applicant (UC-5.10)

3.1.5.11 Receive Pending Documents

Characteristics	Description
Use Case ID	UC-5.11
Use Case Name	Receive Pending Documents
Primary Actors	Registrar Office
Goal	Confirm and update the status of previously missing or pending documents in the system.
Precondition	User must be signed in with Registrar Office credentials.
Scenario	1. Log in to the dashboard. 2. Navigate to the 'Pending Documents' section via the sidebar. 3. Locate the applicant and click the 'OK' or 'Receive' action button.
Exception	Unauthorized access or document verification error.
Frequency of Use	High frequency during the finalization of the admission process.

Table 3.42: Use Case Table: Receive Pending Documents (UC-5.11)

3.1.5.12 View Department Seats Status

3.1.5.13 Export Applicant List

Characteristics	Description
Use Case ID	UC-5.12
Use Case Name	View Department Seats Status
Primary Actors	Registrar
Goal in Context	View Department Seats Status
Precondition	Must be signed in
Triggers	Need to View Department Seats Status
Scenario	1. Log in to dashboard 2. Click department seats status item from sidebar.
Exception	Invalid role
Frequency of Use	High frequency

Table 3.43: View Department Seats Status (UC-5.12)

Characteristics	Description
Use Case ID	UC-5.8
Use Case Name	Export Applicant List
Primary Actors	Registrar Office
Goal	Export the specific applicant data into a spreadsheet or document format.
Precondition	User must be signed in and have access to the applicant records.
Scenario	1. Log in to the dashboard. 2. Navigate to the Applicant table. 3. Select the desired data and click the 'Export' button.
Exception	Unauthorized access or system error during file generation.
Frequency of Use	High frequency for administrative and backup purposes.

Table 3.44: Use Case Table: Export Applicant List (UC-5.8)

3.1.5.14 Create New Record

Characteristics	Description
Use Case ID	UC-5.14
Use Case Name	Create New Record
Primary Actors	Registrar
Goal	Create and download a customized record or list of applicants for office use.
Precondition	User must be signed in with Registrar credentials.
Scenario	1. Log in to dashboard. 2. Click 'Create New Record' from the sidebar. 3. Select required data columns. 4. Provide a file name. 5. Click the 'Download' button to save the record.
Exception	Unauthorized access or invalid file format.
Frequency of Use	High frequency during the processing of applicant data.

Table 3.45: Use Case Table: Create New Record (UC-5.14)

3.1.6 Applicant Profile Management

The Applicant Profile Management module manages the student's interaction with the system, covering everything from registration to document retrieval.

3.1.6.1 Create Account

Characteristics	Description
Use Case ID	UC-6.1
Use Case Name	Create Account
Primary Actors	Applicant
Goal	Create a new account in the admission system.
Precondition	Applicant must be a first-time user and not already registered.
Scenario	1. Provide GST details. 2. Input personal and academic information. 3. Click 'Register' button.
Exception	Invalid GST Applicant ID or data mismatch.

Table 3.46: Use Case Table: Create Account (UC-6.1)

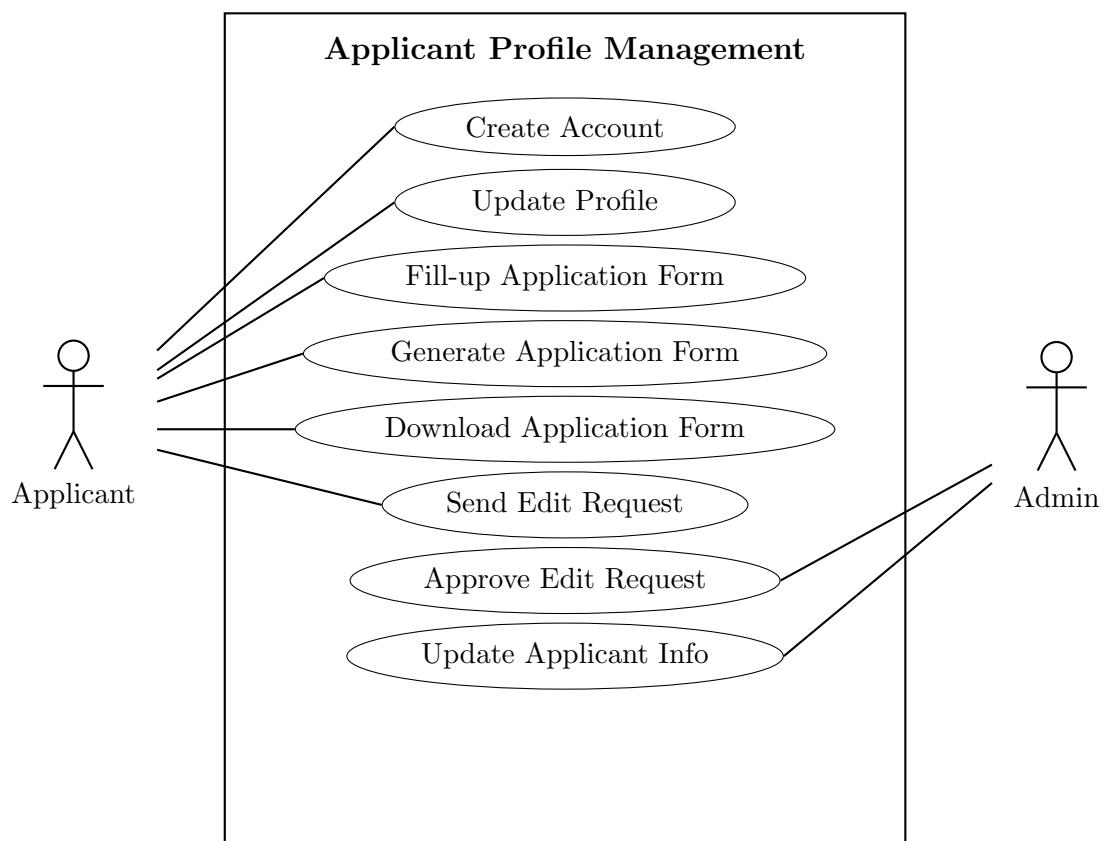


Figure 3.6: Use Case Diagram of Applicant Profile Management

3.1.6.2 Update Profile

Characteristics	Description
Use Case ID	UC-6.2
Use Case Name	Update Profile
Primary Actors	Applicant
Goal	Allow the applicant to modify and keep their personal profile information up to date.
Precondition	Applicant must be successfully logged into their account.
Scenario	1. Visit the Applicant Dashboard. 2. Click on the Profile icon. 3. Select the 'Update Profile' option. 4. Modify the necessary data fields. 5. Click the 'Update' button to save changes.
Exception	Database connection error or session expiration.

Table 3.47: Use Case Table: Update Profile (UC-6.2)

3.1.6.3 Fill-up Application Form

Characteristics	Description
Use Case ID	UC-6.3
Use Case Name	Fill-up Application Form
Primary Actors	Applicant
Goal	Successfully complete and submit the admission application form.
Precondition	Applicant must be signed in to the dashboard.
Scenario	1. Visit Applicant Dashboard. 2. Click 'Fill-up Application Form'. 3. Enter details in all five sequential steps. 4. Review and submit the form.
Exception	Invalid role or session timeout.

Table 3.48: Use Case Table: Fill-up Application Form (UC-6.3)

3.1.6.4 Generate Application Form

Characteristics	Description
Use Case ID	UC-6.4
Use Case Name	Generate Application Form
Primary Actors	Applicant
Goal	Create a downloadable version of the filled application form.
Precondition	Applicant must have completed all five steps of the application form.
Scenario	1. Visit the Applicant Dashboard. 2. Click 'Generate Application Form' button. 3. System compiles data and creates the PDF record.
Exception	Form steps incomplete or invalid user role.

Table 3.49: Use Case Table: Generate Application Form (UC-6.4)

3.1.6.5 Download Application Form

Characteristics	Description
Use Case ID	UC-6.5
Use Case Name	Download Application Form
Primary Actors	Applicant
Goal	Download the admission application form in PDF format for future use.
Precondition	Applicant must be signed in and the form must be generated.
Scenario	1. Visit the Applicant Dashboard. 2. Click 'Download Application Form' button. 3. System serves the PDF file for saving.
Frequency of Use	High frequency

Table 3.50: Use Case Table: Download Application Form (UC-6.5)

3.1.6.6 Send Edit Request

Characteristics	Description
Use Case ID	UC-6.6
Use Case Name	Send Edit Request
Primary Actors	Applicant
Goal	Formally request the Admin for permission to edit information after the application form has been generated.
Precondition	Application form must already be generated and the user must be signed in.
Scenario	1. Visit the Applicant Dashboard. 2. Click the 'Send Edit Request' button. 3. System logs the request for Admin approval.
Frequency of Use	High frequency

Table 3.51: Use Case Table: Send Edit Request (UC-6.6)

3.1.6.7 Approve Edit Request

Characteristics	Description
Use Case ID	UC-6.7
Use Case Name	Approve Edit Request
Primary Actors	Admin
Goal	Grant permission to the applicant to modify their profile data.
Scenario	1. Log in to admin dashboard. 2. Navigate to Edit Requests. 3. Click 'Permit' button to allow applicant update.

Table 3.52: Use Case Table: Approve Edit Request (UC-6.7)

3.1.6.8 Update Applicant Information (Admin)

Characteristics	Description
Use Case ID	UC-6.8
Use Case Name	Update Applicant Information
Primary Actors	Admin
Goal	Allow the Admin to manually correct or update an applicant's profile data in the database.
Precondition	Admin must be authenticated and the specific applicant must exist in the system.
Scenario	1. Log in to Admin dashboard. 2. Navigate to the applicant table. 3. Search for the applicant. 4. Modify the required fields. 5. Click the 'Update' button to save changes via Prisma.
Exception	Invalid user role or database connection error.

Table 3.53: Use Case Table: Update Applicant Information (UC-6.8)

3.1.6.9 User Management

The User Management activity diagram illustrates the workflow of the system's administrative and user functions. Admin and users must first sign in with valid credentials. Once authenticated, the system forks based on user roles and actions.

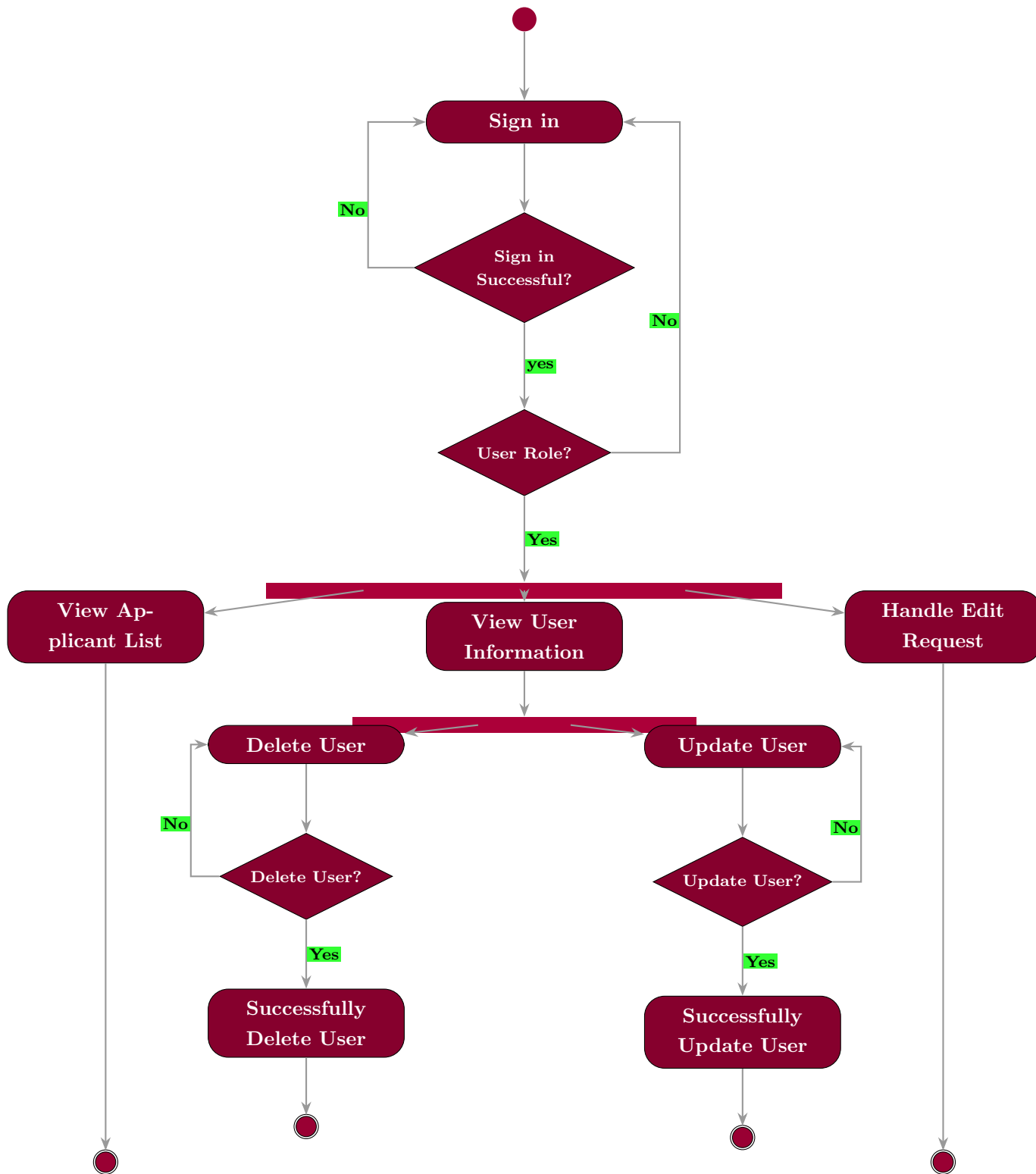


Figure 3.7: Activity Diagram of User Management System

3.1.6.10 Dean Office Management

The Dean Office Management activity diagram illustrates the workflow for faculty-level admission administration. Once the Dean or Registrar signs in successfully, the system allows for diverse operations including applicant searching, document verification, and seat status monitoring.

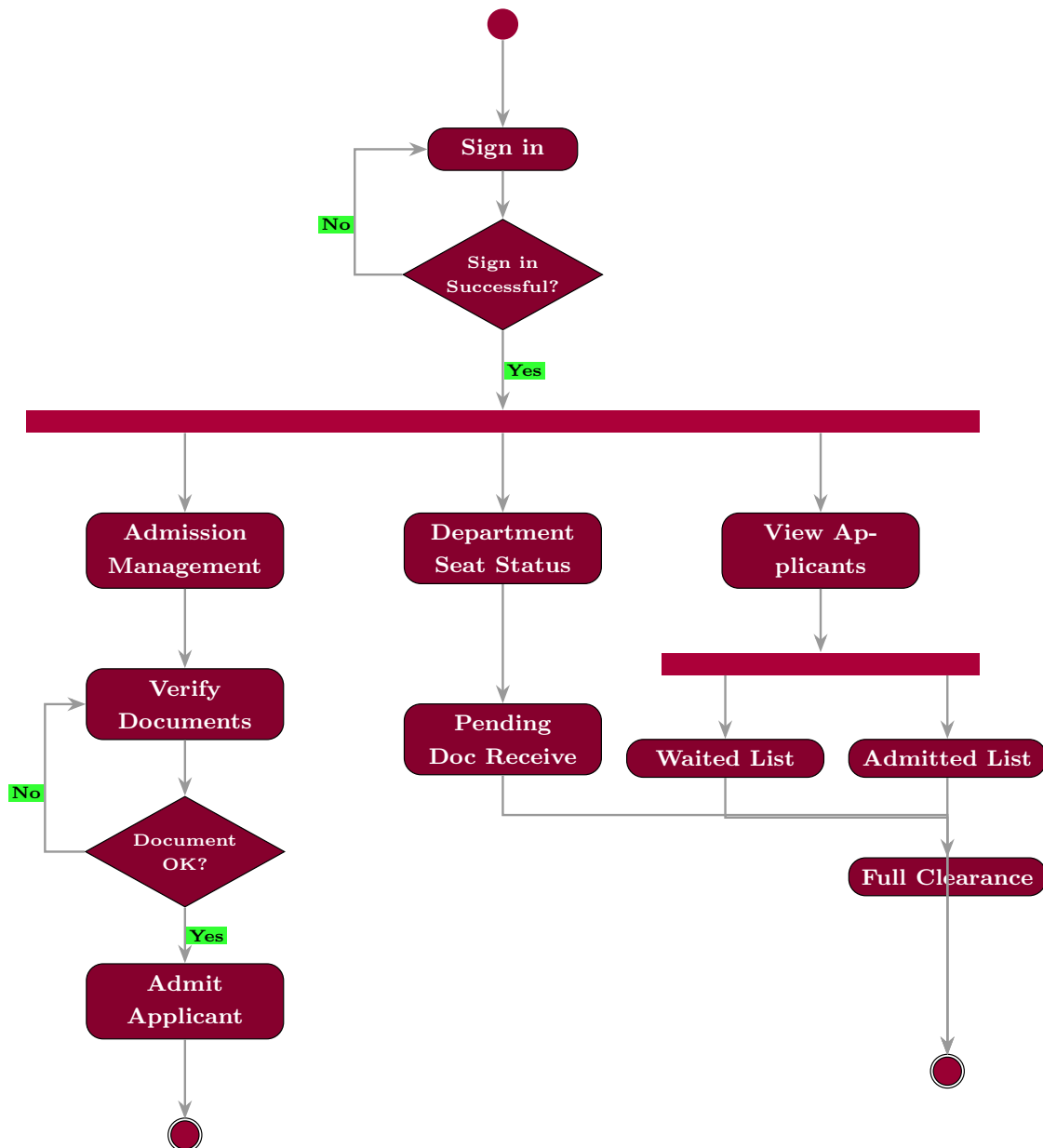


Figure 3.8: Activity Diagram of Dean Office Management (UC-5.0)

3.1.6.11 Hall Office Management

The Hall Office Management activity diagram illustrates the operational workflow for residential hall admission tasks. It covers the process from secure authentication to applicant search, document receipt, and final admission, as well as administrative tasks like exporting applicant lists and downloading official records.

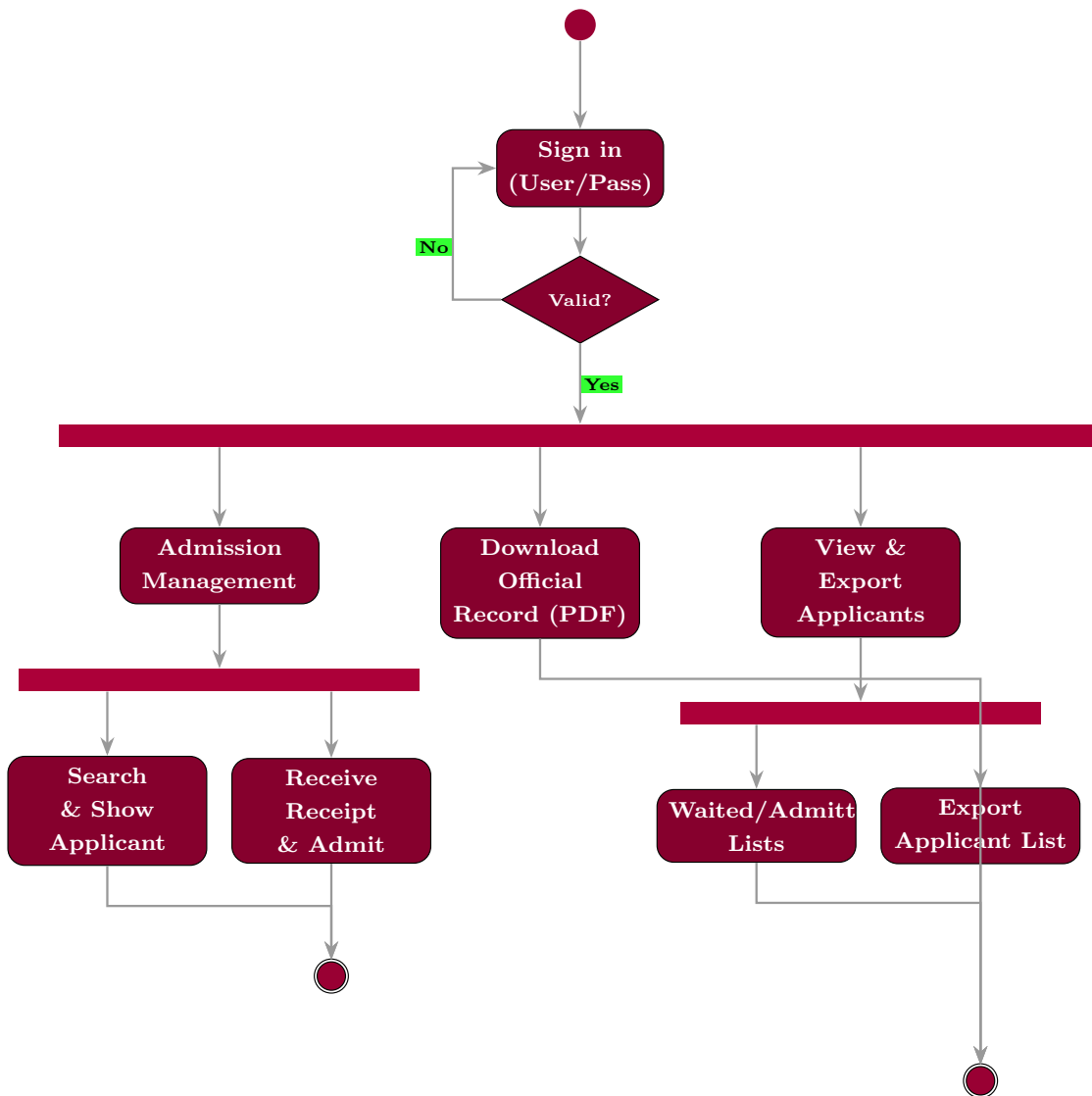


Figure 3.9: Activity Diagram of Hall Office Management (UC-6.0)

3.1.6.12 Medical Office Management

The Medical Office Management activity diagram illustrates the workflow for managing applicant medical records and admissions. After a successful sign-in, the med-

ical officer can search for applicants, update medical information, generate health reports, and manage the admitted and waited applicant lists.

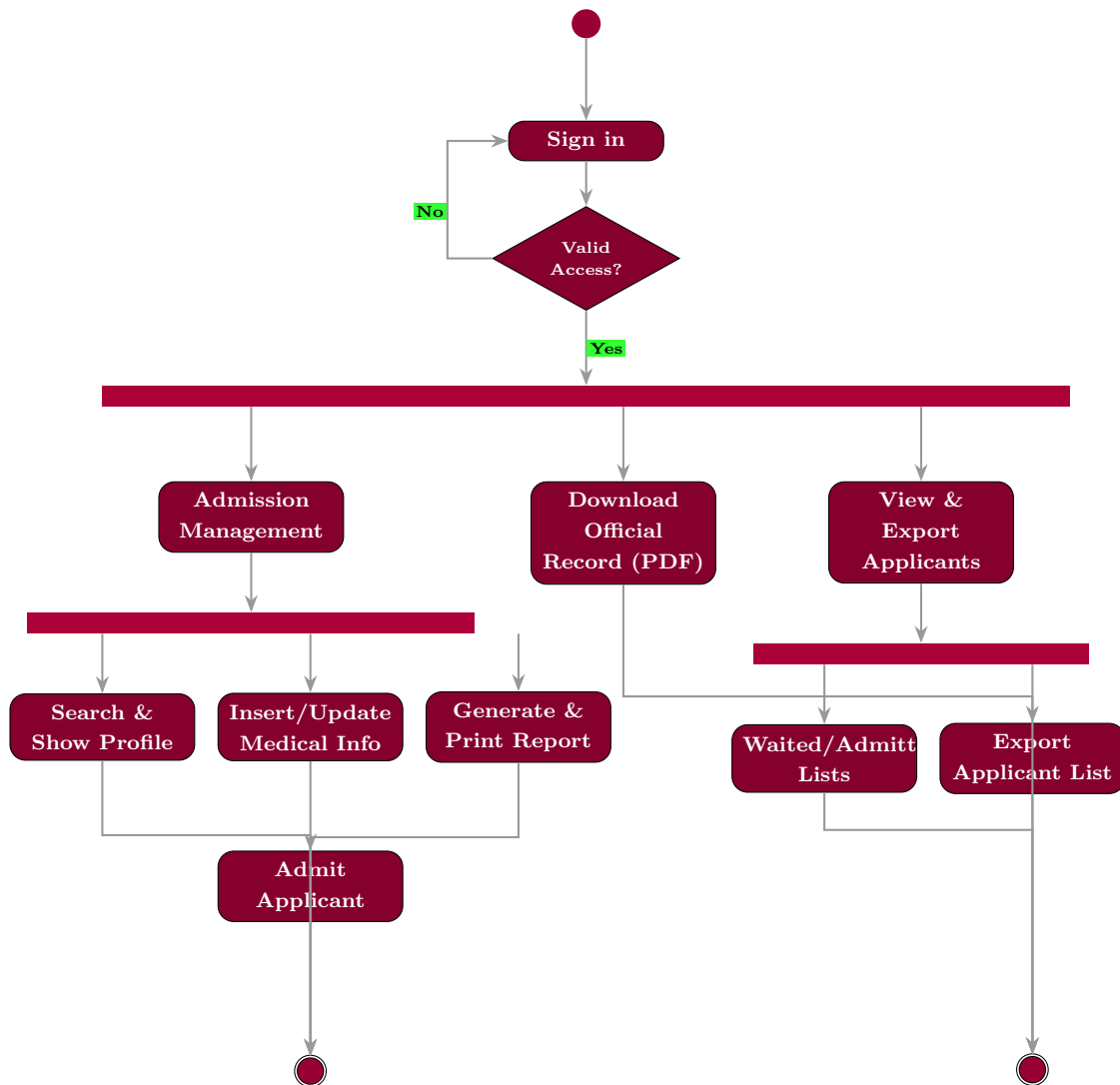


Figure 3.10: Activity Diagram of Medical Office Management (UC-4.0)

3.1.6.13 Registrar Office Management

The Registrar Office Management activity diagram details the high-level administrative tasks handled by the registrar. This includes secure authentication, applicant management (searching, document receipt, and admission), and robust reporting features such as exporting lists and creating customized new records for official documentation.

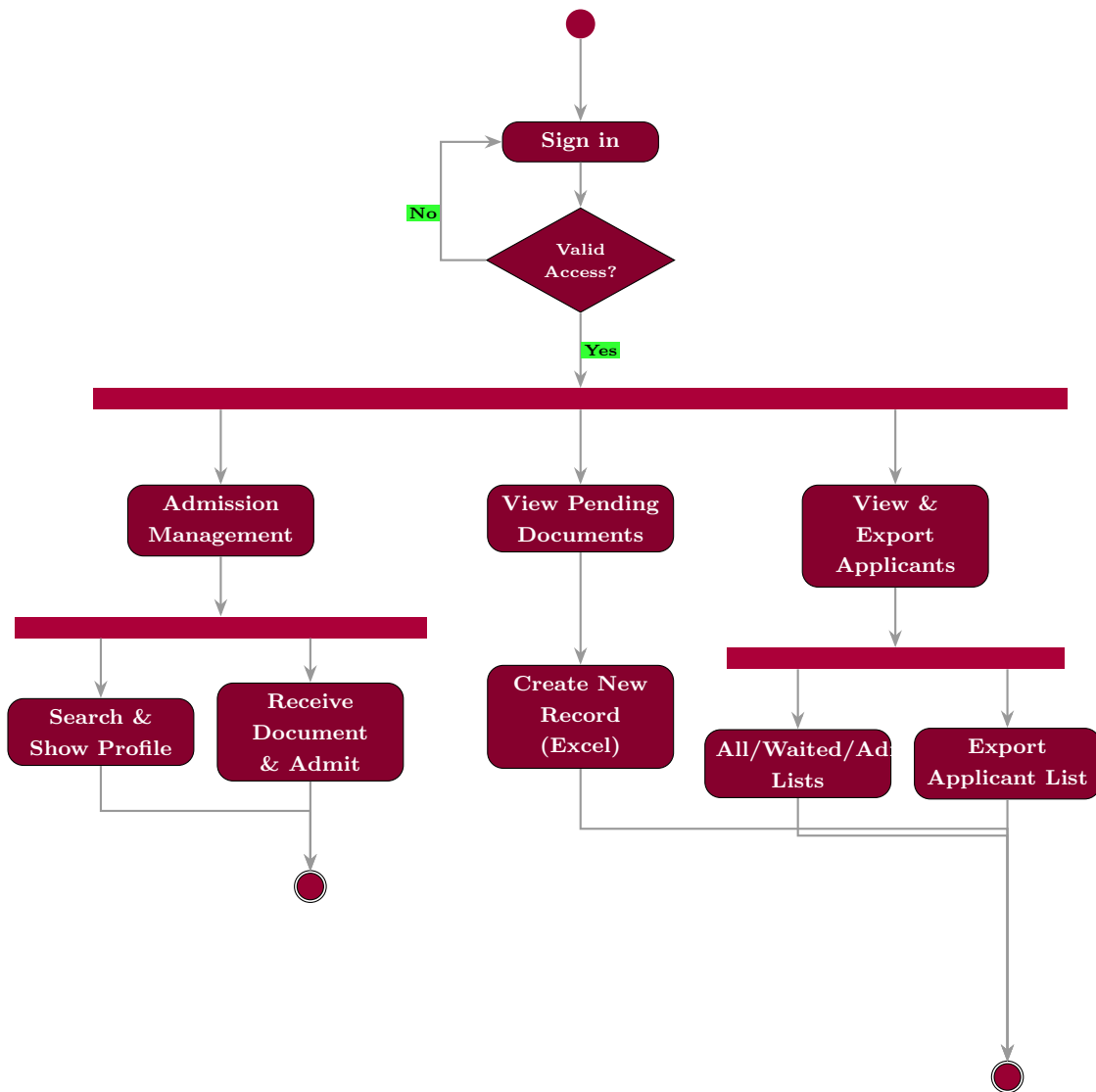


Figure 3.11: Activity Diagram of Registrar Office Management (UC-7.0)

3.1.6.14 Applicant Profile Management

The Applicant Profile Management activity diagram illustrates the interactive workflow for students using the admission system. It encompasses the secure login process using GST credentials, profile updates, the multi-step application form filling process, and the final generation and download of the application form, including the ability to send edit requests post-submission.

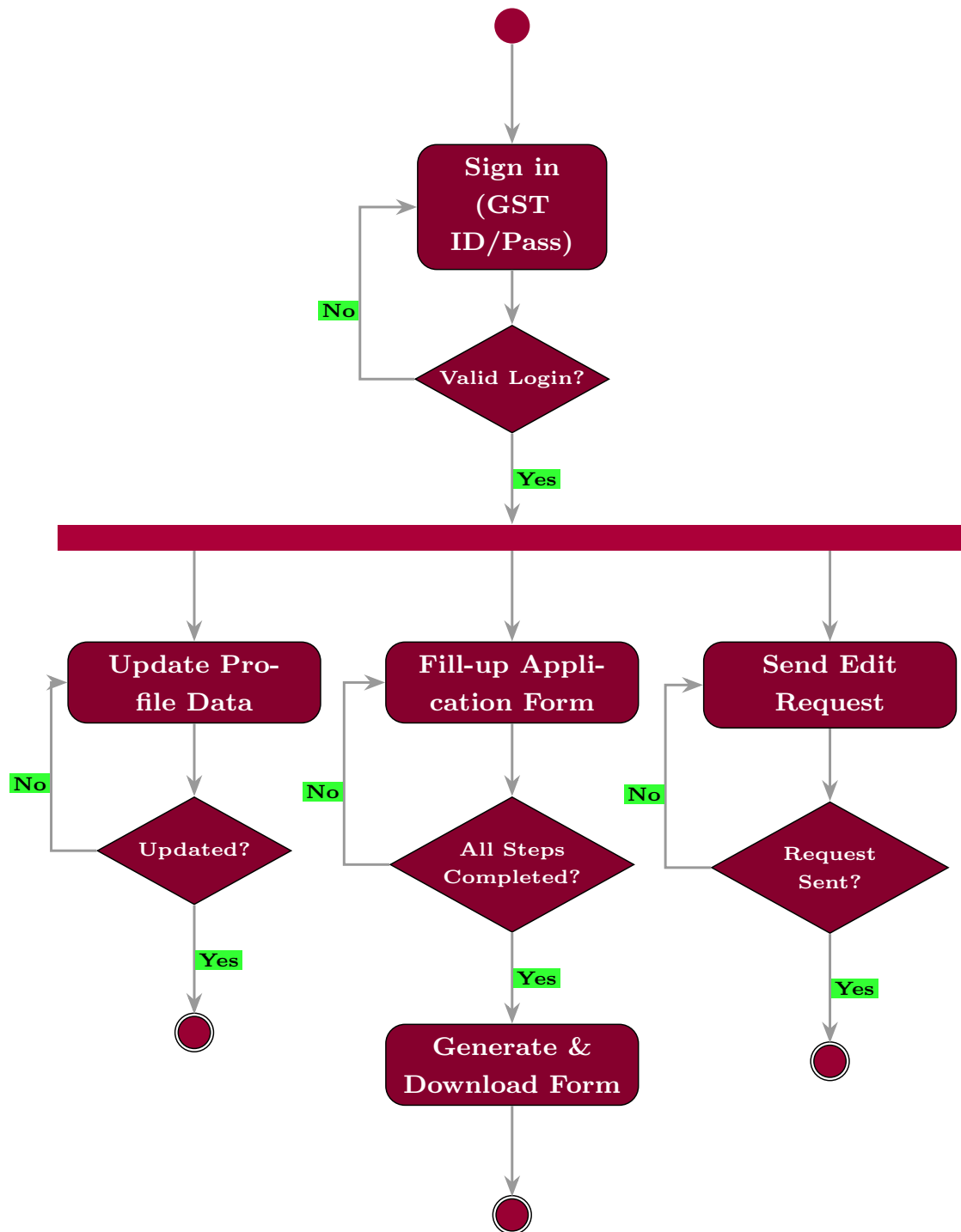


Figure 3.12: Activity Diagram of Applicant Profile Management (UC-2.0)

3.1.7 System Database Schema

The overall database architecture and entity relationships of the JUST Undergraduate Admission System are presented in the following schema diagram. This diagram illustrates the interconnections between user authentication, academic records, and administrative clearance modules.

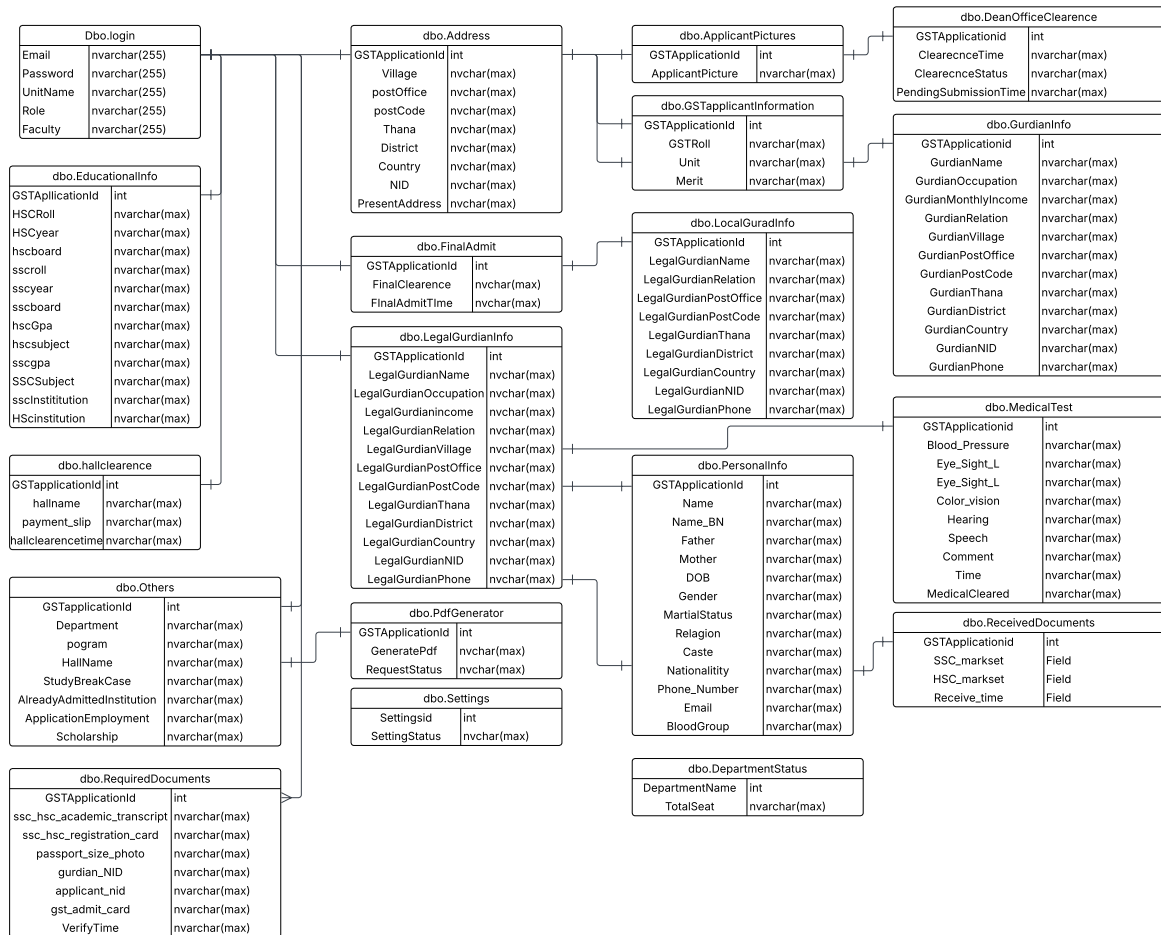


Figure 3.13: Full Database Schema Diagram of the System



Chapter 4

Implementation Details

This chapter discusses the detailed implementation aspects of the JUST Undergraduate Final Admission System (JUST-UFAS). It presents the actual user interfaces developed, focusing on the workflows, data handling, and specific functionalities of different user roles, starting with the authentication process and subsequent role-specific modules.

4.1 Authentication

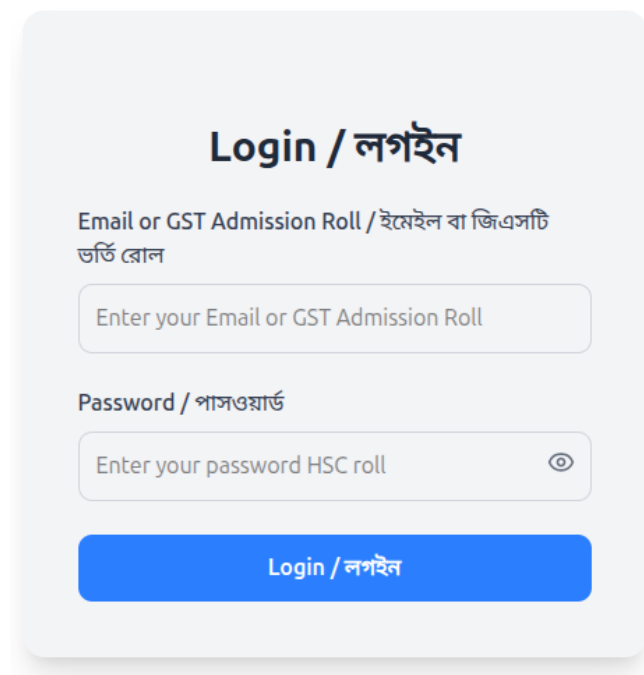
4.1.1 Sign in Page and Process

Authentication is the primary security layer of the system, ensuring that only authorized users can access their respective features. The system provides a centralized login interface for all user roles, including students, administrators, and various office personnel.

The interface is designed with a focus on usability and accessibility, featuring a bilingual (English and Bengali) layout to assist all users. The following section describes the main authentication entry point.

The login page, as shown in Figure 4.1, utilizes a clean and modern card-based design. The authentication workflow involves the following steps:

- **Credential Input:** The user is required to enter a unique identifier and a secure password. For students, the system supports login via their registered **Email** or **GST Admission Roll**. The password field is pre-configured to accept the **HSC Roll** for initial access.
- **Password Visibility:** A toggle icon is integrated into the password field,

A login form titled "Login / লগইন" with a light gray background and rounded corners. It contains two input fields: the first is labeled "Email or GST Admission Roll / ইমেইল বা জিএসটি ভর্তি রোল" and the second is labeled "Password / পাসওয়ার্ড". The second field has a placeholder text "Enter your password HSC roll" and a toggle icon. Below the fields is a blue button labeled "Login / লগইন".

Login / লগইন

Email or GST Admission Roll / ইমেইল বা জিএসটি ভর্তি রোল

Enter your Email or GST Admission Roll

Password / পাসওয়ার্ড

Enter your password HSC roll

Login / লগইন

Figure 4.1: System Authentication (Login Page)

allowing users to verify their input before submission, which reduces entry errors.

- **Submission:** Once the credentials are provided, the user interacts with the **Login** button to initiate the verification process.
- **Redirection:** Upon successful validation, the system's backend determines the user's role and automatically redirects them to the appropriate dashboard (e.g., Student, Dean, or Registrar).
- **Error Handling:** In the event of incorrect credentials, the system prevents access and displays an alert indicating that the username or password does not match, maintaining the security of the application.

4.2 Student Module Implementation

After successful authentication, students are directed to their application portal. The primary task for a student is to complete the comprehensive admission form, providing accurate personal and academic information.

4.2.1 Final Admission Form

The admission form is designed to collect detailed geographical and identification data. To ensure data integrity, the system implements strict front-end validation for each input field.

The form contains the following fields and validation messages:

- গ্রাম / বাড়ি নম্বর / রোড নম্বর / Village / House number / Road number*:**
Input field: Enter Village / House number / Road number
Validation: • Village / House number / Road number is required
- পোস্ট অফিস / Post office*:**
Input field: Enter Post office
Validation: • Post office is required
- পোস্ট কোড / Post code*:**
Input field: Enter Post code
Validation: • Post code must be exactly 4 digits
- থানা / Thana*:**
Input field: Enter Thana
Validation: • Thana is required
- জেলা / District*:**
Input field: Enter District
Validation: • District is required
- দেশ / Country*:**
Input field: Enter Country
Validation: • Country is required
- শিক্ষার্থীর জাতীয় পরিচয়পত্র / জন্ম সনদ / পাসপোর্ট নম্বর / Student's NID / Birth reg. / Passport number*:**
Input field: Enter NID / Birth reg. / Passport number
Validation: • Enter a valid NID, Birth Registration, or Passport number

Figure 4.2: Student Admission Form with Validation

As shown in Figure 4.2, the form focuses on the following components:

- **Address Details:** The form requires the student's *Village/House number*, *Post Office*, *Thana*, *District*, and *Country*. Each field is marked with an asterisk (*), indicating it is mandatory.
- **Postal Code Validation:** A specific validation rule is applied to the **Post Code** field, requiring exactly 4 digits, which prevents incorrect data entry.
- **Identification Information:** The system provides a flexible identification section where students must enter either their **NID**, **Birth Registration**, or **Passport number**.
- **Real-time Error Handling:** As illustrated in the figure, the system uses reactive validation. If a user leaves a field empty or enters invalid data (e.g., an incomplete post code), a red error message appears immediately below the input field to guide the user.

- **Bilingual Interface:** Continuing the design philosophy, all labels are provided in both Bengali and English to ensure clarity for all applicants.

This structured approach ensures that the database receives clean, formatted data, which simplifies the subsequent verification process by the Dean and Registrar offices.

4.2.2 Application Summary and PDF Download

After completing the necessary steps of the admission form, the student reaches the final stage where they can review their application summary and export the data for official use.

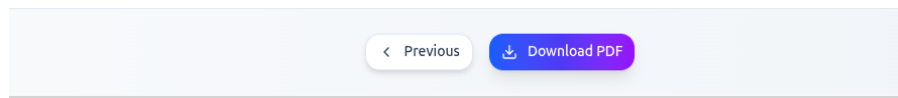


Figure 4.3: Final Application Submission and Download Interface

The features of this section, as shown in Figure 4.3, are described below:

- **Review Mechanism:** The *Previous* button allows students to navigate back and double-check their provided information before final submission. This ensures that no incorrect data is sent to the university database.
- **Instant PDF Generation:** The system is integrated with a PDF generation library that converts the student's input fields into a formal document format. By clicking the **Download PDF** button, the user can instantly save a copy of their admission form.
- **Visual Feedback:** The button utilizes a distinct gradient color and a download icon to provide clear visual feedback, indicating that this is the primary action of the page.
- **Official Requirement:** This downloaded PDF acts as an acknowledgment slip, which contains a unique identifier for the student. It is mandatory for students to bring a printed copy of this document during the physical verification at the Dean's office.

4.2.3 Submission Confirmation and Complaint Management

Once a student successfully submits their application form, the system displays a confirmation message and provides options for subsequent necessary actions. This interface ensures the applicants that their data has been securely stored in the database.

The screenshot shows a web interface for form submission confirmation. At the top, there's a header section with the title "ভর্তি ফর্ম সাবমিশন" (Form Submission) and a subtitle "আপনার তথ্য যাচাই করুন এবং প্রয়োজনীয় পদক্ষেপ গ্রহণ করুন" (Verify your information and take necessary steps). Below this, there's a green alert box with the text "সফলভাবে সম্পন্ন হয়েছে" (Successfully completed) and "আপনার তথ্য সফলভাবে সংরক্ষণ করা হয়েছে। নিচে থেকে PDF ডাউনলোড করতে পারেন।" (Your information has been successfully saved. You can download PDF from below). Underneath, there's a section titled "পরবর্তী পদক্ষেপ" (Next Steps) containing two buttons: a blue button labeled "PDF ডাউনলোড করুন" (Download PDF) and a white button labeled "অভিযোগ করুন" (Complain).

Figure 4.4: Form Submission Confirmation and Complaint Interface

As illustrated in Figure 4.4, the submission status page includes several key functional elements:

- **Success Notification:** A green alert box is utilized to inform the student that their information has been successfully saved. This enhances the user experience and maintains system transparency.
- **PDF Generation and Download:** Students can download their completed application form in PDF format for future reference. This is a core component of the system's automated report generation module.
- **Complaint Management:** If a student notices any discrepancy in their data or encounters issues after submission, they can directly file a report through this integrated module.
- **Action Buttons:** The interface features clear and actionable buttons (e.g., "Download PDF" and "Complain"), which guide the user through the next steps without ambiguity.

Through this feature, system reliability is increased, and students are provided with immediate feedback and support options.

4.3 Admin Module: Data and Resource Management

The Admin module serves as the primary gateway for preparing the system for a new admission session.

4.3.1 Bulk Data Import and External Storage Integration

The administrator can populate the database by uploading Excel or SQL files containing GST admission results. To ensure data safety, a Google Drive integration is also provided.

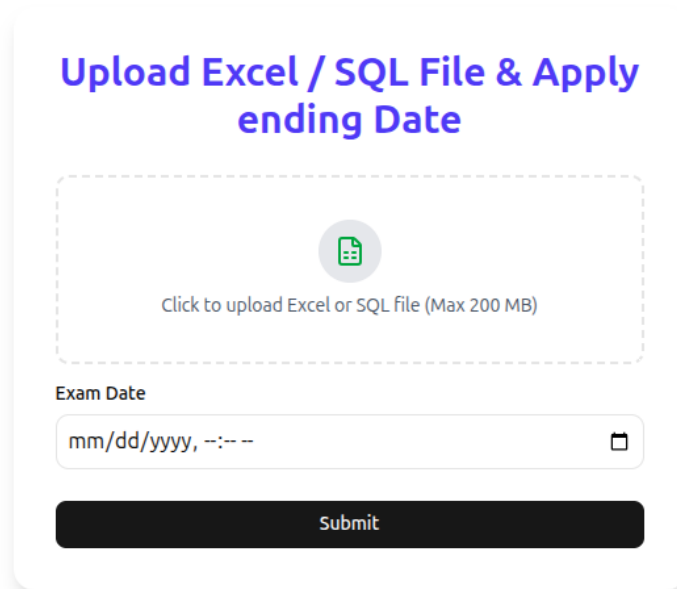
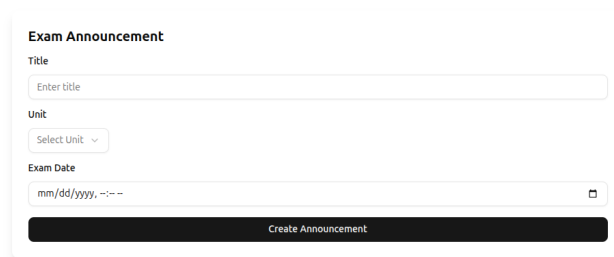


Figure 4.5: Excel/SQL File Upload and Application Deadline Interface

- **Data Ingestion:** The system parses the uploaded file and updates the applicant records. The interface also allows setting an "Exam Date" as a deadline.
- **Cloud Backup:** Using the interface shown in *uploadDrive.png* (if applicable), the admin can sync local data with Google Drive for disaster recovery.

4.3.2 Examination Announcement

The admin defines the specific admission unit and the corresponding examination date to notify the applicants.



The image shows a web form titled "Exam Announcement". It contains three input fields: "Title" with a placeholder "Enter title", "Unit" with a dropdown menu labeled "Select Unit", and "Exam Date" with a date picker showing "mm/dd/yyyy, --:--". At the bottom is a black button labeled "Create Announcement".

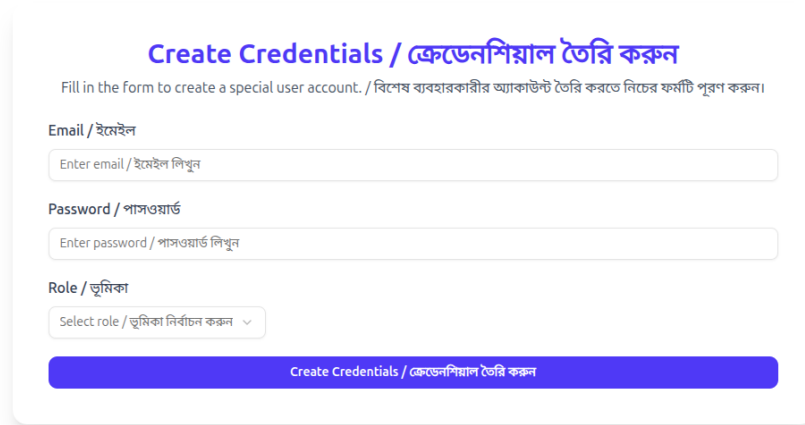
Figure 4.6: Exam Announcement Configuration Interface

4.4 Admin Module: User Control and Security

Security is managed through a comprehensive user administration panel that controls access for university officials.

4.4.1 Role-Based Credential Generation

The Admin can create specific accounts for Deans, Registrars, and Medical Officers using a dedicated credential creation form.



The image shows a web form titled "Create Credentials / ক্রেডেনশিয়াল তৈরি করুন". Below the title is a subtitle in Bengali: "Fill in the form to create a special user account. / বিশেষ ব্যবহারকারীর অ্যাকাউন্ট তৈরি করতে নিচের ফর্মটি পূরণ করুন।". The form has three sections: "Email / ইমেইল" with a text input field "Enter email / ইমেইল লিখুন", "Password / পাসওয়ার্ড" with a text input field "Enter password / পাসওয়ার্ড লিখুন", and "Role / ভূমিকা" with a dropdown menu "Select role / ভূমিকা নির্বাচন করুন". At the bottom is a blue button labeled "Create Credentials / ক্রেডেনশিয়াল তৈরি করুন".

Figure 4.7: User Credential Creation Interface

4.4.2 System User Monitoring

A centralized dashboard allows the Admin to monitor all registered officials, their roles, and their account status.

EMAIL	ROLE	STATUS	CREATED AT	ACTION
medical@gmail.com	MEDICAL	ACTIVE	2026-04-08T05:43:50.453Z	Block
hallregister@gmail.com	HALL_REGISTER	ACTIVE	2026-04-08T05:39:32.256Z	Block
register@gmail.com	REGISTER	ACTIVE	2026-04-08T05:39:12.274Z	Block
dean@gmail.com	DEAN	ACTIVE	2026-04-08T05:38:20.729Z	Block
admin@gmail.com	ADMIN	ACTIVE	2026-03-31T16:20:55.058Z	Block
admingmail.com	ADMIN	ACTIVE	2026-03-31T16:04:31.818Z	Block

Figure 4.8: Administrative User Management Dashboard

- **Status Control:** Admins can "Block" or "Unblock" accounts instantly to prevent unauthorized access.
- **Role Identification:** The interface clearly labels the role (e.g., DEAN, REGISTER) and the timestamp of account creation.

4.5 Admin Module: Complaint Resolution System

To address student grievances, the Admin panel provides a real-time tracking system for complaints submitted through the student portal.

4.5.1 Reviewing Pending Complaints

All student issues are listed in a simplified queue, allowing administrators to address them systematically.

Pending Complaints

GST APPLICATION ID	STATUS	ACTION
123456	Pending	View

Figure 4.9: Pending Complaint Tracking Interface

The complaint handling workflow involves:

- **Application ID Mapping:** Every complaint is mapped to a GST Application ID for cross-verifying academic records.
- **Detailed Action:** By clicking the "View" button, the admin can read the full text of the complaint and take necessary corrective measures.

4.6 Dean Module: Applicant Verification

The Dean's primary responsibility is to oversee the departmental applicant lists and verify the academic eligibility of the candidates.

4.6.1 Departmental Applicant Queue

The portal provides a real-time data table displaying students assigned to various departments under the faculty.

All Applicants

[Export Excel](#)

GST APPLICATION ID	UNIT	NAME	DEPARTMENT	HSC BOARD	HSC ROLL	HSC YEAR	OMR PHYSICS	OMR CHEMISTRY	OMR MATH	OMR BIOLOGY
123986	A	Sovo	N/A	BARISHAL	123456	2022	13	15.75	3.75	16.75
123466	A	Saniul	N/A	BARISHAL	123456	2022	13	15.75	3.75	16.75
123459	A	Ramjan	N/A	BARISHAL	123456	2022	13	15.75	3.75	16.75
123458	A	Tamim	N/A	BARISHAL	123456	2022	13	15.75	3.75	16.75
123457	A	Rakesh	N/A	BARISHAL	123456	2022	13	15.75	3.75	16.75

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Figure 4.10: Dean's Portal: Integrated Applicant List

4.6.2 Detailed Information Table

To facilitate precise verification, a detailed list table is implemented that shows specific student attributes.

GST APPLICATION ID	UNIT	NAME	DEPARTMENT	HSC BOARD	HSC ROLL	HSC YEAR	CHANGE DEPARTMENT
123456	A	Nafis	Computer Science and Engineering	BARISHAL	123456	2022	Computer Science and Engineering

Figure 4.11: Dean’s Detailed Data Verification Table

The Dean uses this interface to cross-check GST scores and departmental requirements before moving to the approval stage.

4.7 Dean Module: Student Status Management

The Dean holds the authority to either approve a student for a department or cancel the application based on academic discrepancies.

GST APPLICATION ID	UNIT	NAME	DEPARTMENT	HSC BOARD	HSC ROLL	HSC YEAR	REGISTER APPROVED
123456	A	Nafis	Computer Science and Engineering	BARISHAL	123456	2022	Approved

Figure 4.12: Dean’s Portal: Admission Approval and Cancellation Interface

The Dean can perform the following actions:

- **Approve Admission:** If the student’s marks and documents are valid, the Dean clicks the ”Approve” button to move the applicant to the next stage.
- **Cancel Admission:** In case of fraudulent data or failing to meet departmental criteria, the Dean has the provision to cancel the admission status immediately.

4.8 Dean Module: Data Redundancy and Sync

To ensure the security of academic records, the Dean’s module is equipped with an automated data synchronization feature.

4.8.1 Google Drive Integration

The system allows the Dean to sync the verified applicant lists directly to a faculty-specific Google Drive account.

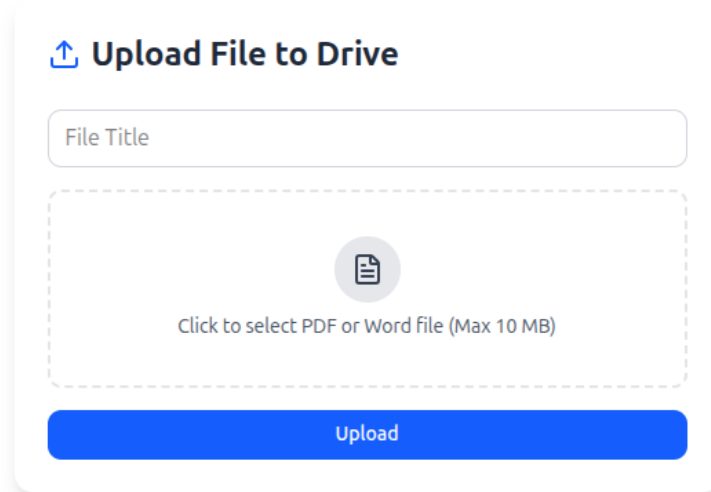


Figure 4.13: Faculty Data Synchronization with Cloud Storage

This integration serves as a disaster recovery measure, ensuring that the Dean’s office maintains a permanent record of all admitted students even in the event of local server issues.

4.9 Registrar Module: Centralized Data Validation

The Registrar office holds the authority for the final validation of applicant data before the admission is officially confirmed.

4.9.1 Master Applicant Queue

The Registrar monitors the global applicant list to oversee the progress of the admission session across all faculties.

All Applicants

All ▾
Q Search...
Export Excel

GST APPLICATION ID	UNIT	NAME	DEPARTMENT	HSC BOARD	HSC ROLL	HSC YEAR	OMR PHYSICS	OMR CHEMISTRY	OMR MATH	OMR BIOLOGY
123986	A	Sovo	N/A	BARISHAL	123456	2022	13	15.75	3.75	16.75
123466	A	Saniul	N/A	BARISHAL	123456	2022	13	15.75	3.75	16.75
123459	A	Ramjan	N/A	BARISHAL	123456	2022	13	15.75	3.75	16.75
123458	A	Tamim	N/A	BARISHAL	123456	2022	13	15.75	3.75	16.75
123457	A	Rakesh	N/A	BARISHAL	123456	2022	13	15.75	3.75	16.75

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Figure 4.14: Registrar’s Master Applicant Data Interface

4.9.2 Detailed Verification Matrix

Using a more granular data table, the Registrar office cross-checks the physical document submission status with the digital records.

All ▾
Q Search...
Export Excel

GST APPLICATION ID	UNIT	NAME	DEPARTMENT	HSC BOARD	HSC ROLL	HSC YEAR	CHANGE DEPARTMENT
123456	A	Nafis	Computer Science and Engineering	BARISHAL	123456	2022	Computer Science and Engineering ▾

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Figure 4.15: Registrar’s Detailed Validation Table

This dual-layer verification ensures that only students with legitimate academic credentials and fee payments are processed for the final step.

4.10 Registrar Module: Data Archiving and Redundancy

As the primary record-keeper of the university, the Registrar’s module emphasizes permanent data storage and cloud synchronization.

4.10.1 Automated Cloud Backup

To prevent data loss and ensure long-term accessibility, the system integrates a direct upload feature to the university's central Google Drive.

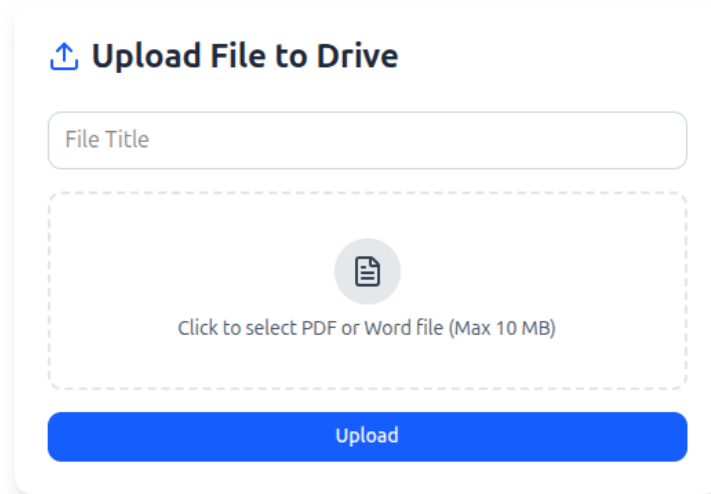


Figure 4.16: Registrar's Archival System (Google Drive Sync)

Through this interface, the Registrar can archive the entire session's data, providing a robust disaster recovery plan for the admission system.

4.11 Registrar Module: Final Authorization

The Registrar performs the final audit. This interface allows the Registrar to finalize the admission or reject it if any legal or official requirement is missing.

All

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GST APPLICATION ID	UNIT	NAME	DEPARTMENT	HSC BOARD	HSC ROLL	HSC YEAR	REGISTER APPROVED
123456	A	Nafis	Computer Science and Engineering	BARISHAL	123456	2022	Approved

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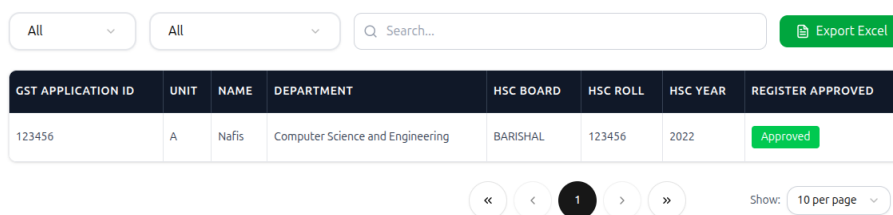
Figure 4.17: Registrar's Final Admission Authorization Dashboard

Through this dashboard, the Registrar ensures:

- **Final Confirmation:** Authorized students are officially enrolled in the university database.
- **Admission Rejection:** The Registrar can cancel or revert an admission if the payment or official verification fails at the last moment.

4.12 Hall Registrar: Residential Clearance

The Hall Registrar manages the residential status. Only students with an "Approved" hall status are allowed to reside in the university halls.



GST APPLICATION ID	UNIT	NAME	DEPARTMENT	HSC BOARD	HSC ROLL	HSC YEAR	REGISTER APPROVED
123456	A	Nafis	Computer Science and Engineering	BARISHAL	123456	2022	Approved

Figure 4.18: Hall Allocation Approval and Cancellation View

- **Seat Approval:** Assigning a hall name and room to the student.
- **Allocation Cancellation:** Removing a student from the hall list if they choose to stay off-campus or provide incorrect information.

4.13 Hall Registrar Module: Resident Data Verification

To ensure accurate record-keeping, the Hall Registrar utilizes a specialized queue to manage applicants who have applied for residential facilities.

4.13.1 Hall Applicant Queue

The interface displays a list of students who have cleared the academic and administrative stages and are now waiting for residential allocation.

All Applicants

All ▾
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Export Excel

GST APPLICATION ID	UNIT	NAME	DEPARTMENT	HSC BOARD	HSC ROLL	HSC YEAR	OMR PHYSICS	OMR CHEMISTRY	OMR MATH	OMR BIOLOGY
123986	A	Sovo	N/A	BARISHAL	123456	2022	13	15.75	3.75	16.75
123466	A	Saniul	N/A	BARISHAL	123456	2022	13	15.75	3.75	16.75
123459	A	Ramjan	N/A	BARISHAL	123456	2022	13	15.75	3.75	16.75
123458	A	Tamim	N/A	BARISHAL	123456	2022	13	15.75	3.75	16.75
123457	A	Rakesh	N/A	BARISHAL	123456	2022	13	15.75	3.75	16.75

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Figure 4.19: Hall Resident Applicant Queue

4.13.2 Detailed Resident Attributes

The Hall Registrar can view specific student data, such as permanent address and gender, which are crucial factors in hall assignment.

All ▾
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Export Excel

GST APPLICATION ID	UNIT	NAME	DEPARTMENT	HSC BOARD	HSC ROLL	HSC YEAR	CHANGE DEPARTMENT
123456	A	Nafis	Computer Science and Engineering	BARISHAL	123456	2022	Computer Science and Engineering ▾

« < 1 > »
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Figure 4.20: Detailed Resident Information Matrix

4.14 Hall Registrar Module: Data Redundancy

Due to the sensitivity of residential records, the Hall Registrar module includes a dedicated synchronization mechanism to preserve allocation data.

4.14.1 Cloud Synchronization for Hall Records

The Hall Registrar can sync the final residential allotment lists to a secure cloud platform, ensuring that hall offices have access to student data for room distribution.

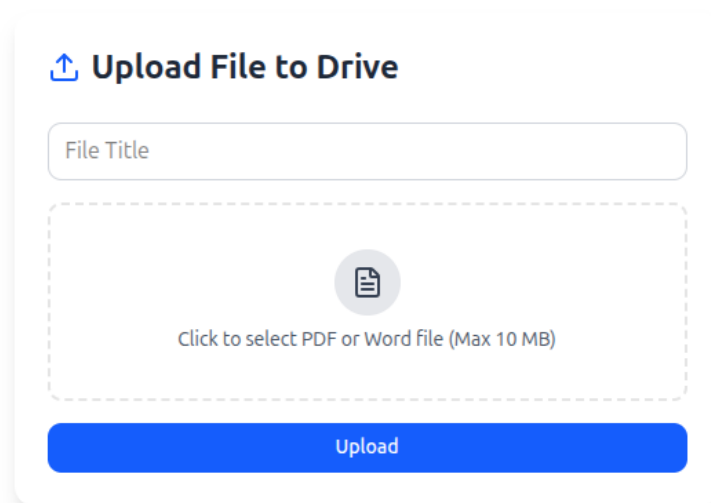


Figure 4.21: Hall Office Data Sync and Backup

This feature ensures data integrity and provides an offline reference for hall administration personnel.

4.15 Hall Registrar: Residential Clearance

The Hall Registrar manages the residential status. Only students with an "Approved" hall status are allowed to reside in the university halls.

All

All

Q Search...

Export Excel

GST APPLICATION ID	UNIT	NAME	DEPARTMENT	HSC BOARD	HSC ROLL	HSC YEAR	REGISTER APPROVED
123456	A	Nafis	Computer Science and Engineering	BARISHAL	123456	2022	Approved

«

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Figure 4.22: Hall Allocation Approval and Cancellation View

- **Seat Approval:** Assigning a hall name and room to the student.
- **Allocation Cancellation:** Removing a student from the hall list if they choose to stay off-campus or provide incorrect information.

4.16 Medical Officer Module: Applicant Health Data

This section allows the Medical Officer to access the list of applicants who are awaiting medical checkups or record verification.

4.16.1 Medical Verification Queue

The portal displays a prioritized list of students who have completed their initial academic formalities and are now eligible for medical screening.

All Applicants

All

Q Search...

Export Excel

GST APPLICATION ID	UNIT	NAME	DEPARTMENT	HSC BOARD	HSC ROLL	HSC YEAR	OMR PHYSICS	OMR CHEMISTRY	OMR MATH	OMR BIOLOGY
123986	A	Sovo	N/A	BARISHAL	123456	2022	13	15.75	3.75	16.75
123466	A	Saniul	N/A	BARISHAL	123456	2022	13	15.75	3.75	16.75
123459	A	Ramjan	N/A	BARISHAL	123456	2022	13	15.75	3.75	16.75
123458	A	Tamim	N/A	BARISHAL	123456	2022	13	15.75	3.75	16.75
123457	A	Rakesh	N/A	BARISHAL	123456	2022	13	15.75	3.75	16.75

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Figure 4.23: Medical Officer's Applicant Screening Queue

4.16.2 Detailed Medical Record Matrix

A specialized table is used to view detailed student attributes, helping the medical staff to identify and document any specific health conditions.

All

Q Search...

Export Excel

GST APPLICATION ID	UNIT	NAME	DEPARTMENT	HSC BOARD	HSC ROLL	HSC YEAR	CHANGE DEPARTMENT
123456	A	Nafis	Computer Science and Engineering	BARISHAL	123456	2022	Computer Science and Engineering

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Figure 4.24: Detailed Health Record and Information Matrix

4.17 Medical Officer Module: Data Synchronization

To ensure the long-term safety of student health records, a cloud synchronization feature is integrated into the Medical Module.

4.17.1 Cloud Backup for Health Records

The Medical Officer can sync all health-related data and clearance reports to the university's central Google Drive for future medical history tracking.

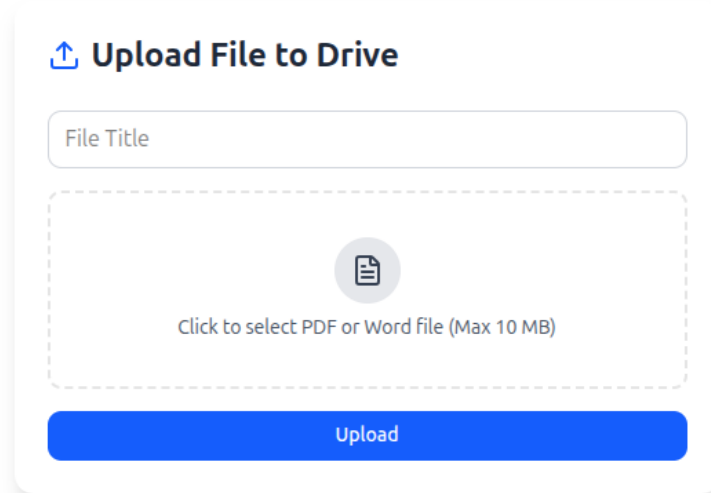


Figure 4.25: Medical Data Archiving and Drive Integration

This integration ensures that the university medical center maintains a persistent and secure digital record of all admitted students.

Chapter 5

Testing and Quality Assurance

Software testing is an indispensable phase of the development life cycle for the JUST Undergraduate Final Admission System (JUST-UFAS). It ensures the reliability, security, and correctness of the system before it is deployed for the 2026 admission session. This chapter details the multi-layered testing strategy employed, ranging from automated unit tests to manual functional verification.

5.1 Automated White Box Testing with Vitest

As the system is built on the Next.js framework, we utilized **Vitest** for high-performance unit testing. White box testing allows us to verify the internal logic, functions, and components of the application.

5.1.1 Component Rendering Tests

Using *Vitest* alongside *React Testing Library*, we ensured that the core UI components render without crashing.

- **Verification Point:** We tested the rendering of the landing page, navigation bar, and admission form.
- **Outcome:** The tests confirmed that the "JUST-UFAS" title and essential action buttons are correctly mounted in the DOM.

5.1.2 Logical and Utility Function Testing

We implemented unit tests for independent functions that handle sensitive data logic.

- **Date Formatting:** A utility function was tested to ensure that ISO strings from the database are converted into user-friendly formats (e.g., "DD-MM-YYYY") for admission deadlines.
- **GPA Calculation:** Functions responsible for validating GST scores against department-specific eligibility criteria were tested with various edge cases.

5.2 Backend API Testing with Postman

To ensure the robustness of our RESTful APIs developed with Node.js and Prisma, we conducted extensive testing using **Postman**.

- **Endpoint Integrity:** We sent various HTTP requests (GET, POST, PATCH) to endpoints like `/api/applicants` and `/api/auth`.
- **JSON Schema Validation:** Verified that the backend returns the correct JSON structure. For instance, the applicant list API must return an array of objects containing valid GST IDs and merit positions.
- **Status Code Verification:** Ensured that the server returns appropriate status codes: 200 OK for success, 401 Unauthorized for invalid logins, and 400 Bad Request for malformed data.

5.3 Manual Functional Testing

Manual testing was performed to simulate real-world user behavior and ensure the system's usability from a student and administrator's perspective.

5.3.1 Form Validation Testing

The admission form is the most critical part of the student module. We performed exhaustive manual tests on all input fields.

- **Empty Field Validation:** Attempting to submit the form without mandatory fields (e.g., Permanent Address) triggers an immediate error message via Tailwind-styled toast notifications.
- **Input Constraints:** We tested numeric fields to ensure that character inputs are restricted, preventing potential database errors.

5.3.2 Role-Based Access Control (RBAC) Test

We manually verified that the security boundaries between different user roles are strictly enforced.

- **Verification:** A student account was used to attempt access to the Dean’s approval dashboard (/admin/dean).
- **Result:** The system successfully redirected the user to the login page, proving that the middleware protection is functioning correctly.

5.4 Summary of Test Results

The overall testing phase concluded with a 100% pass rate for all critical modules. Table 5.1 summarizes the key test outcomes.

Table 5.1: Detailed Testing Outcome Summary

Test Category	Testing Tool	Objective	Status
Unit Testing	Vitest	Internal logic validation	Passed
API Testing	Postman	REST endpoint integrity	Passed
UI Validation	Manual	Form errors & UX	Passed
Security (RBAC)	Manual/Postman	Unauthorized access prevention	Passed
Data Integrity	Prisma/SQL	CRUD operation verification	Passed

Chapter 6

Deployment and Conclusion

The final phase of the software development life cycle is the deployment and delivery of the system to the production environment. This chapter discusses the deployment strategy used for the JUST Undergraduate Final Admission System (JUST-UFAS) and the future prospects of the project.

6.1 Deployment Process

Deploying a modern web application built with Next.js and Node.js involves a structured workflow to ensure stability and performance. The deployment process for JUST-UFAS followed these essential steps:

- **Production Build:** The application code was optimized and compiled into a production-ready build to ensure fast loading times and efficient resource management.
- **Environment Configuration:** Essential environment variables, such as PostgreSQL database connection strings and Prisma client settings, were configured on the production server.
- **Server Setup:** The application is hosted on the university's official infrastructure. Due to existing resource allocations, the system is currently deployed on a specific port.
- **Maintenance and Monitoring:** The system is actively maintained by the **Technical Committee of Undergraduate Admission** at Jashore University of Science and Technology.

6.2 Deployment Status and URL

The system is currently live and accessible for the final admission process.

- **Current Hosting:** The project is running on the university's existing domain using port 3000 as a cost-effective hosting solution.

6.3 Conclusion

The "JUST Undergraduate Final Admission System" offers a significant improvement over manual admission processes. By automating the workflow, the system has achieved:

- **Efficiency:** Faster data processing and reduced manual labor.
- **Accuracy:** Minimized human errors in student data handling and merit list verification.
- **Cost-Effectiveness:** Reduced need for extra manpower, leading to operational cost savings for the university.

6.4 Future Work

While the current system fulfills all primary requirements, future updates could include:

- **Automated Migration:** Implementing a full auto-migration system for students across different departments.
- **Payment Gateway Integration:** Integrating local mobile banking (bKash, Nagad) for seamless admission fee collection.
- **Mobile Application:** Developing a dedicated mobile app for students to track their admission status in real-time.

Through this project, Jashore University of Science and Technology has taken a step forward in digitizing its administrative infrastructure, ensuring a smoother experience for future students.